

# Applied Microeconomics — Master Notes

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by Jakob V. Stangel

One running document for the whole course, organised by lecture (= textbook chapter). Format per lecture: **Résumé** → ★ **Heart of it** → **Key concepts** → **Section-by-section** → **Cases** → **Key terms** → **Exam pointers**. Textbook: Richard Friberg, *A Free Introduction to Microeconomics* · Lecturer: Luigi Butera (lbu.eco@cbs.dk). Exam: written sit-in, *open book, No AI*, Week 50 (9 Dec 2026).

## Overview — where everything is

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(Everything below is clickable.)

- **Lecture 1: Principles of Economics (Ch. 1)** — what is economics? · the three principles · opportunity cost & marginal thinking · models: positive vs normative · ★ Fun facts
- **Lecture 2: Supply and Demand (Ch. 2)** — demand & the demand function · supply & the supply function · ★ solving market equilibrium · shifts & comparative statics · price ceilings & floors · ★ Fun facts
- **Lecture 3: Elasticities & Taxes (Ch. 2 recap · §6.4)** — why elasticity? · ★ price elasticity of demand · computing & the point-elasticity trick · elastic / inelastic / unit-elastic · cross-price · income elasticity & Engel curves · supply elasticity · elasticity over time · ★ taxes: the wedge, incidence & equivalence · formula sheet · ★ practice problems (with solutions) · ★ Fun facts
- **Lecture 4: Consumer Choice (Ch. 5)** — modelling choice: the 3 ingredients · preferences & rationality axioms · utility & utility functions · ★ marginal utility · indifference curves & their 3 types · ★ marginal rate of substitution (MRS) · budget constraint & MRT · ★ the optimum:  $MRS = MRT$  · comparative statics · interior vs corner · Cobb-Douglas shortcut · behavioral economics · cases · formula sheet · ★ practice problems (with solutions) · ★ Fun facts

## Lecture 1: Principles of Economics (Chapter 1)

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**Required reading:** Friberg, ch. 1 (*Introduction*).

**Theme of the lecture (Butera):** the "economic way of thinking" — economics studies how people allocate scarce resources among competing needs, using **three principles** (optimization · equilibrium · empiricism) and **models** (simplified, testable) to understand and predict behaviour.

★ **The heart of the chapter:** economics = doing the best you can with scarce resources. The toolkit is **three principles** — **Optimization** · **Equilibrium** · **Empiricism**; the two workhorses of optimization are **opportunity cost** and **marginal thinking**; and the method is building **positive (testable) models**, not making normative judgments. "*All models are useful, but all models are wrong.*"

**The red thread.** Everything starts from **scarcity** — without it we wouldn't need economics. From scarcity flows: you must **choose** (optimize) → every choice has an opportunity cost and is decided **at the margin** → many people choosing interact until no one wants to move (**equilibrium**) → and we test our claims about all this with **models + data** (empiricism). For any decision ask: *what's scarce · what's the best alternative given up (opportunity cost) · what does one more unit give me (marginal) · is this claim positive (testable) or normative (a value judgment)?*

## Key concepts / "modes" to use

- **Scarcity** — limited resources vs unlimited wants; the reason economics exists.
- **The three principles: Optimization · Equilibrium · Empiricism.**
- Opportunity cost — the value of the *best alternative* you give up.
- Marginal thinking — decide by the value of *one more* unit; law of diminishing marginal returns.
- **Model** — a simplified representation of relationships between variables ("all models are wrong, but useful").
- **Positive vs normative** — a *testable* cause-effect claim vs a *value judgment*.

## Section-by-section main points

### | What is economics?

**Core:** economics = **the study of how people allocate scarce resources over competing needs** — "people (try to) do the best they can for themselves with what they have."

- **Scarcity of two things:** the **things we want** (goods, services, jobs, education, relationships, recognition...) *and* the **means to get them** (money, **time, attention**, information, patience, willpower). Air to breathe isn't scarce → not an economic question.
- Every decision is shaped by **scarcity** (can't have it all), **other people** (you're not an island), and **the rules of the game** (you can't do whatever you want).
- **How societies allocate: central planners** (someone at the top decides who gets what — parents, government) vs **markets** (value set by supply & demand — from job markets to dating markets). Economics gives tools to predict how we decide and when markets do (or don't) allocate *efficiently*.

### | The three principles of economics

**Core:** the whole course rests on three ideas.

| Principle             | Meaning                                                                                                                        |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| ① <b>Optimization</b> | We choose the <b>best feasible option</b> given our limited resources (money, time, attention, willpower are all scarce).      |
| ② <b>Equilibrium</b>  | A situation in which <b>no one would benefit</b> from changing their <i>own</i> behaviour, given the choices of everyone else. |
| ③ <b>Empiricism</b>   | We build <b>models</b> → models generate <b>predictions</b> → we use <b>data/experiments</b> to test (falsify) them.           |

## Optimization — opportunity cost + marginal thinking

**Core:** to optimise, use two tools — opportunity cost (weigh each use against its best alternative — "apples to apples") and marginal thinking (what does *one more* unit give me?).

Opportunity cost = the value of the best alternative you give up — nothing "free" is free if it uses a scarce resource.

### OPPORTUNITY COST

$$OC(\text{action}) = \text{benefit}(\text{action}) - \text{value of the best alternative given up}$$

*Reads:* the true cost of a choice is what it gives you **minus** what you gave up to get it. — **OC(action)** = the opportunity cost of doing it • **benefit(action)** = the gain the action delivers • **value of the best alternative given up** = the payoff of the *next-best* option you sacrificed by choosing this one.

- *Social media:* 1 hr/day on TikTok isn't free — you give up sport, study, work. Monetise the time with the hourly wage (avg **118 DKK** in Denmark):

### YEARLY TIME-COST OF 1 HR/DAY ON TIKTOK

$$118 \text{ DKK/hr} \times 1 \text{ hr/day} \times 365 \text{ days} = \mathbf{43,070 \text{ DKK / year}}$$

*Reads:* price the time you *could* have been earning, added up over a year. — **118 DKK/hr** = Denmark's average hourly wage (i.e. the value of one hour of your time) • **1 hr/day** = the time you spend • **365 days** = a full year → **43,070 DKK** of wages forgone.

- *Trip to Berlin (worked example):* drive = 1,280 DKK & 14 h; fly = 1,930 DKK & 3 h. Flying frees **11 h** you could work:

### OPPORTUNITY COST OF DRIVING INSTEAD OF FLYING

$$OC(\text{drive}) = (1,930 - 1,280) - (118 \times 11) = 650 - 1,298 = \mathbf{-648 \text{ DKK}} \Rightarrow \mathbf{\text{fly}}$$

*Reads:* the money you save by driving, minus the value of the time driving wastes. — **(1,930 - 1,280)** = the extra *ticket* cost of flying (650 DKK) • **(118 × 11)** = the value of the **11 hours** flying saves you, priced at the 118 DKK/hr wage (1,298 DKK) • a **negative** result means driving is **648 DKK worse** than flying → **fly**.

*(Still a model — what if you enjoy the drive?)*

**Marginal thinking:** decide by the value of the **next** unit — keep going while marginal benefit exceeds marginal cost; stop where they meet. That point is the **optimum**:

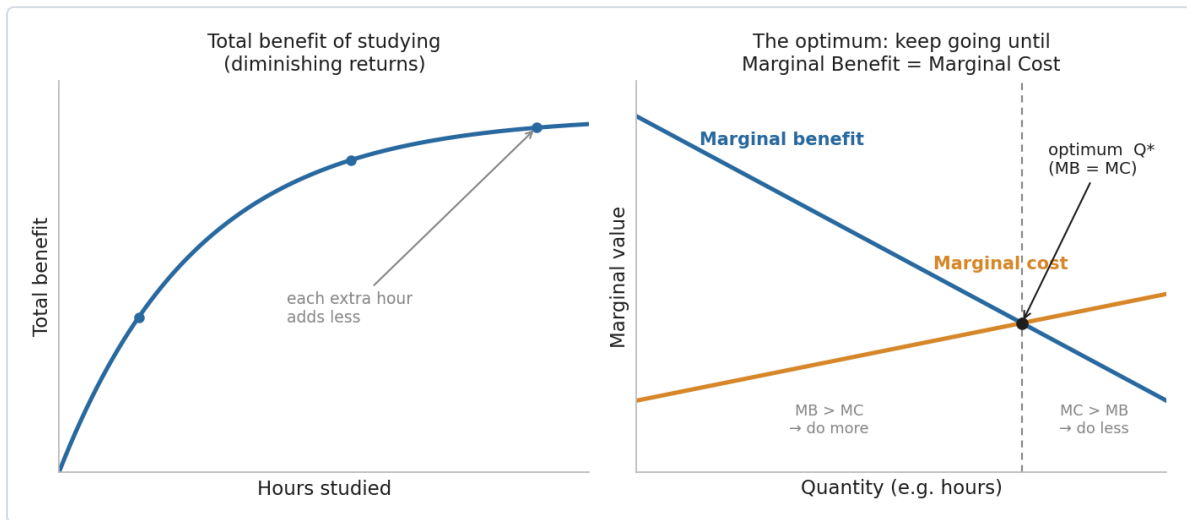
### OPTIMUM CONDITION

$$\text{choose the quantity where } \mathbf{\text{Marginal Benefit} = \text{Marginal Cost}} \quad (\text{MB} = \text{MC})$$

*Reads:* keep going until one more unit's benefit exactly equals its cost. — **Marginal Benefit (MB)** = the *extra* benefit from **one more** unit • **Marginal Cost (MC)** = the *extra* cost of that unit • at the **optimum**

$MB = MC$  (while  $MB > MC$  do more; once  $MC > MB$  you've gone too far).

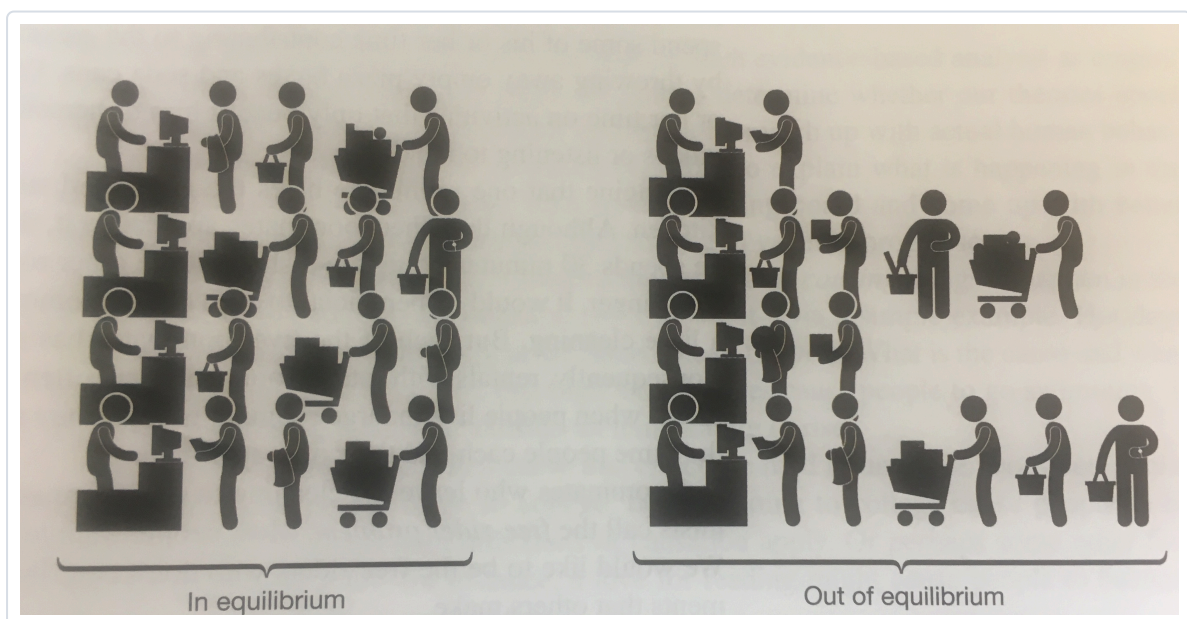
- *Night before the exam*: total benefit of studying rises then flattens; the **marginal benefit of each extra hour falls** (law of diminishing marginal returns). Optimal hours = where MB of one more hour just equals its MC.



How to read it (illustrative diagram — the graph the book draws in a later chapter). *Left*: total benefit rises but each extra hour adds less — the curve flattens (diminishing returns). *Right*: that flattening means **marginal benefit slopes down**; **marginal cost slopes up**. Left of the crossing,  $MB > MC$  so one more unit is worth it (*do more*); right of it,  $MC > MB$  (*do less*). The best you can do is the quantity where they meet — the optimum  $Q^*$ , where  $MB = MC$ .

## Equilibrium

**Core**: an **equilibrium** is a resting point — given what everyone else is doing, **no individual can make themselves better off by changing their own behaviour**. Market prices are the classic example (later lectures): they adjust until quantity supplied = quantity demanded and no one wants to move.



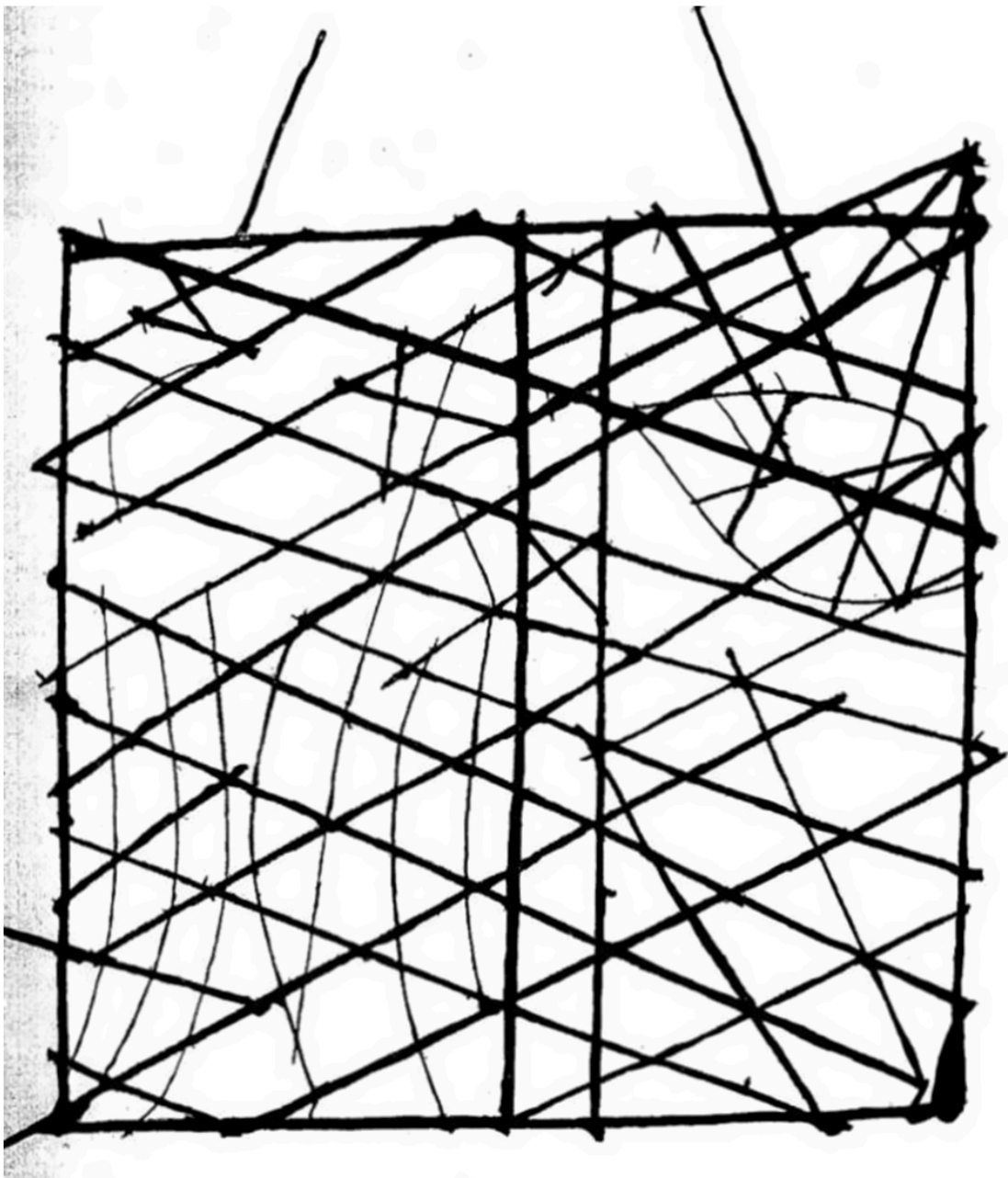
How to read it (**equilibrium** = supermarket queues, from the lecture). Each lane is a checkout; the queue length is how many people are waiting. **Out of equilibrium** (right) the lanes are uneven, so

anyone in a long line can still gain by switching to a shorter one — people keep moving. **In equilibrium** (left) all lanes are equal, so **no one can do better by changing lane on their own** — that "no incentive to move" resting point is exactly what equilibrium means (and why market prices settle where supply meets demand).

## Empiricism — models, and positive vs normative

**Core:** economics reasons with **models** and tests them against **data** — and it deals in **positive** (testable) statements, not **normative** (value) ones.

- A **model** expresses relationships between variables and makes **simplifying assumptions** (abstracts from reality). "*All models are useful, but all models are wrong*" — the art is choosing the right model (Keynes).



**How to read it (Friberg Fig. 1.1 — what a "model" is).** This is a Polynesian **stick chart**: sticks map the ocean's swell patterns, shells mark islands. It's not a literal picture of the Pacific — it **strips reality down to just the features you need to navigate**. That's the book's metaphor for an economic model: deliberately simplified, "wrong" as a full description, yet **good enough to steer decisions by**.

- **Positive vs normative:**
  - **Positive** = a **testable/falsifiable** cause-effect claim — "*each additional Starbucks in a ZIP code is associated with a 0.5% rise in housing prices*" (Glaeser et al. 2018).
  - **Normative** = a **value judgment** — "*gentrification is bad because it forces the poor out.*"
  - **Economics is about positive statements**, not normative ones.
- **Correlation  $\neq$  causation:** in summer people eat more ice-cream *and* drown more — ice-cream doesn't cause drowning (a hidden cause: heat). A theory is only useful if it's **empirically falsifiable**.
- **Expanding models to fit data:** "people only give to charity because they care about others' well-being" fails the data (public spending doesn't crowd out giving 1-for-1 — Andreoni 1990; people give to *inefficient* charities). So we add "**joy of giving**" (utility from the act of giving itself).

## **Friberg's framing — allocation mechanisms (Ch. 1's example)**

**Core:** a scarce good (e.g. **university places**) can be allocated many ways — and each has different effects on **efficiency** and **equity**. This is the book's opening illustration of "the role of economics."

- **Non-price mechanisms:** aptitude tests/interviews • secondary-school grades • open entry + hard exams • **waiting lists** • **queues** ("camp on the sidewalk") • **lotteries** • social class/legacy.
- **Price mechanisms:** everyone pays the **same fee** (set high enough to clear demand) • **auction** places to the highest bidders.
- Real systems **combine** several (fees + tests + interviews). Micro gives the tools to analyse the **trade-offs** (who gets in, how tied to parents' income, effect on effort).

## **Cases & examples**

Each: *the example* → *the concept it teaches*.

- **1 hr/day of TikTok** → opportunity cost ( $\approx 43,070$  DKK/yr at the average wage) — nothing using scarce time is free.
- **Drive vs fly to Berlin** → opportunity cost done properly: include the *value of time*, and driving turns out to *cost* 648 DKK more.
- **Studying the night before an exam** → marginal thinking + diminishing returns; optimise where  $MB = MC$ .
- **Ice-cream & drowning** → **correlation  $\neq$  causation**.
- **Starbucks & house prices** (Glaeser 2018) → a **positive** statement; **gentrification "is bad"** → a **normative** one.
- **Charitable giving** (Andreoni 1990) → models must be **expanded to fit data** ("joy of giving").
- **University admissions** (Friberg) → **allocation mechanisms** (price vs non-price) and their equity/efficiency trade-offs.

## Key terms

| Term                                | Meaning                                                           |
|-------------------------------------|-------------------------------------------------------------------|
| Scarcity                            | Limited resources vs unlimited wants — the basis of economics     |
| Optimization                        | Choosing the best feasible option given constraints               |
| Equilibrium                         | No one gains from changing their own behaviour, given others'     |
| Empiricism                          | Build models → predict → test with data                           |
| Opportunity cost                    | Value of the best alternative given up                            |
| Marginal thinking                   | Deciding by the value of one more unit                            |
| Law of diminishing marginal returns | Each extra unit adds less than the last                           |
| Model                               | Simplified representation of relationships between variables      |
| Positive statement                  | A testable/falsifiable cause-effect claim                         |
| Normative statement                 | A value judgment (good/bad, should/ought)                         |
| Allocation mechanism                | Rule that decides who gets a scarce good (price/lottery/queue...) |

## Exam pointers

- State the **three principles** (Optimization · Equilibrium · Empiricism) and define each.
- **Compute an opportunity cost** properly (include the value of time) — the Berlin example is the model answer.
- Use **marginal thinking** +  $MB = MC$  to find an optimum, and name the **law of diminishing marginal returns**.
- Classify a statement as **positive vs normative**, and remember **economics = positive**; watch for **correlation  $\neq$  causation**.
- Explain why economists use **models** ("all models are wrong, but useful") and give an **allocation-mechanism** trade-off (Friberg's university example).

## ★ Fun facts & memorable details

Sticky lines from Lecture 1.

- Keynes: "*Economics is a science of thinking in terms of models joined to the art of choosing models which are relevant to the contemporary world.*"
- The course mantra: "*All models are useful, but all models are wrong.*"
- **1 hour of TikTok a day  $\approx$  43,070 DKK a year** in forgone wages (at Denmark's  $\sim$ 118 DKK average hourly wage).

- Driving to Berlin *looks* cheaper (1,280 vs 1,930 DKK) but, once you price the 11 lost hours, it actually **costs 648 DKK more** than flying.
- "During the summer people eat more ice-cream and also drown more" — the classic **correlation-≠-causation** trap (the hidden cause is heat).
- "Joy of giving" (Andreoni 1990): we don't donate only for others' welfare — we get utility from the *act* of giving, which is why public spending doesn't crowd out private giving 1-for-1.
- Learning micro is "*a first course in a new language*" — a bit hard, not immediately intuitive, needs practice.

## Lecture 2: Supply and Demand (Chapter 2)

Required reading: Friberg, ch. 2 (*Supply and Demand*). • Lecturer: Luigi Butera.

**Theme of the lecture (Butera):** build tractable models of demand and supply, then use them to find the **market equilibrium** — the price at which "*no one would benefit from changing their own behaviour given the choices of others.*" The whole of micro is understanding that equilibrium: what sets prices and quantities, and how they move.

★ **The heart of the chapter:** a market is one picture — a **downward-sloping demand curve** and an **upward-sloping supply curve** that cross at **exactly one point, the equilibrium ( $p^*$ ,  $q^*$ )**, where **quantity demanded = quantity supplied**. Three skills: ① write demand & supply as **functions**; ② set them equal to **solve for  $p^*$  and  $q^*$** ; ③ **shift** a curve and read off the new equilibrium. *Price is set by BOTH blades of the scissors (Marshall) — never demand or supply alone.*

**The red thread.** For *every* question ask the one diagnostic that trips people up: **is this a *movement* ALONG a curve (only the good's own price changed) or a *SHIFT* of the whole curve (something else changed — income, other prices, costs, tastes, tech)?** Own price → slide along. Anything else → the curve moves. Get that right and the rest is algebra.

### Key concepts / "modes" to use

- **Demand function**  $Q_D = D(p, p_s, p_c, Y, \tau)$  → its 2-D demand curve; **inverse demand** ( $p$  as a function of  $q$ ).
- **Supply function**  $Q_S = S(p, p_o)$  → **supply curve**; **inverse supply**.
- **Movement along vs shift** of a curve (the core distinction); **shifters** of each curve.
- **Substitutes / complements**; **normal / inferior** goods; **Giffen** good (the rare exception).
- **Market equilibrium**  $Q_D = Q_S$  → solve for  $p^*$ ,  $q^*$ ; **excess demand (shortage)** / **excess supply (surplus)**.
- **Comparative statics** (shifts) & how the **slope** of the other curve sizes the effect.
- **Price ceiling / floor**; **perfect competition** assumptions; **Marshall's scissors**.

### Section-by-section main points

## 2.1 Demand — the demand function

**Core:** demand answers "how much do buyers want at each price?" The **law of demand:** all else equal, **lower price → higher quantity demanded** (the curve slopes **down**).

- *In plain terms — why it slopes down:* two reasons. ① **Scarcity** — you don't have infinite money, so a lower price lets you afford more. ② **Diminishing marginal utility** — the *first* slice of pizza is worth a lot to you; the 30th almost nothing, so you'll only buy more if it's cheaper.
- **The demand function** describes quantity demanded  $Q_D$  as a function of price **and everything else**:

### DEMAND FUNCTION

$$Q_D = D(p, p_s, p_c, Y, \tau)$$

*Reads:* how much buyers want depends on the good's own price **and** a handful of other factors. —  $Q_D$  = quantity demanded (what we're solving for) •  $D(\cdot)$  = "is a function of" •  $p$  = the good's **own price** •  $p_s$  = price of a **substitute** •  $p_c$  = price of a **complement** •  $Y$  = consumers' **income** •  $\tau$  = **taxes**.

- **How the 2-D curve appears** — "flush" the other variables into a constant. A demand *curve* plots  $Q_D$  against **own price only**, so you **fix** everything else at set numbers; they collapse into the intercept. The book's worked reduction — start from a specific form and plug in temperature  $T=20^\circ$ , substitute price  $p_r=2$ , income  $I=5$ :

### COLLAPSING TO A LINEAR DEMAND CURVE

$$Q_D = 1 - p + 0.25 \cdot T + 0.75 \cdot p_r + 0.5 \cdot I = 1 - p + 5 + 1.5 + 2.5 = 10 - p$$

(intercept  $a = 10$  bundles all the non-price factors)

*Reads:* plug fixed numbers into every non-price factor and they merge into one constant. —  $T$  = temperature ( $=20$ ) •  $p_r$  = price of raspberries ( $=2$ ) •  $I$  = income ( $=5$ ); the **coefficients** (0.25, 0.75, 0.5) say how strongly each shifts demand • adding the constant pieces ( $1 + 5 + 1.5 + 2.5$ ) gives the **intercept  $a = 10$** , so only  $p$  is left as a variable →  $Q_D = 10 - p$ .

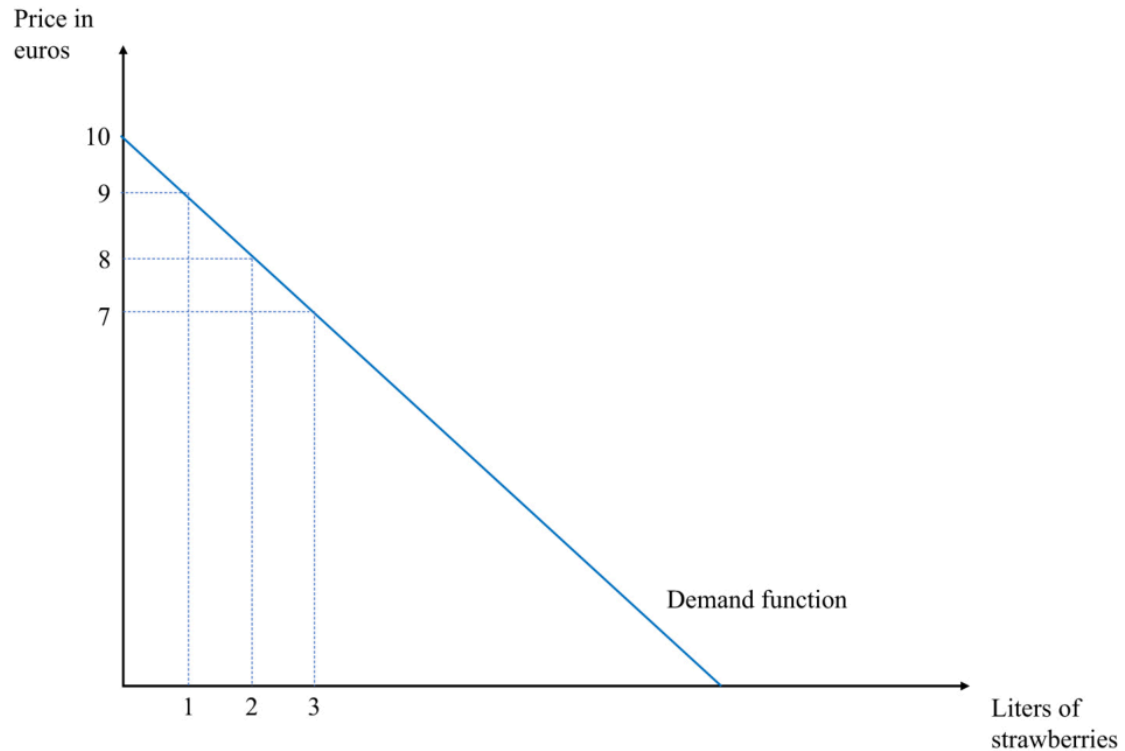
### LINEAR DEMAND & ITS INVERSE (FOR GRAPHING)

$$Q_D = a - b \cdot p \rightarrow \text{e.g. } Q_D = 10 - p \text{ (} a = 10, b = 1 \text{)}. \text{ Solve for } p \Rightarrow \text{inverse demand: } p = 10 - Q_D \cdot \text{(deck's version: } Q_D = 20 - 2p \Rightarrow p = 10 - \frac{1}{2} \cdot Q_D \text{)}$$

*Reads:* a straight-line demand — quantity falls as price rises. —  $a$  = the **intercept** (quantity demanded if price were 0; it bundles income, other prices, tastes — *shift it and the whole curve moves*) •  $b$  = the **slope** (how many units demand drops per +1 in price) •  $p$  = price. The **inverse** just rearranges it to put  $p$  alone on the left, because graphs put price on the vertical axis.

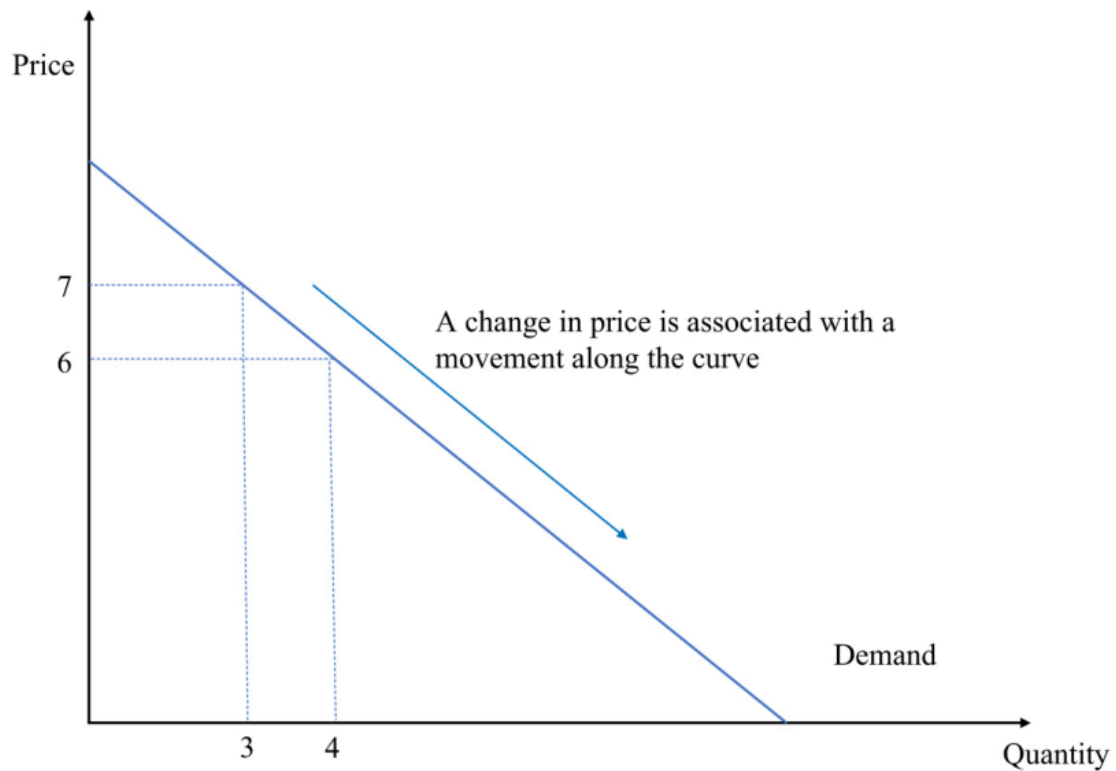
*Check:*  $p = 7 \rightarrow Q_D = 3$ ;  $p = 6 \rightarrow Q_D = 4$ .  $a$  = the intercept (shift it and the whole curve moves);  $b$  = how many units demand falls per +1 in price.

## Supply and demand



**How to read it (the demand curve).** By convention **price is on the vertical axis**, quantity on the **horizontal** (blame Alfred Marshall). Each point answers *both* questions: "at price  $p$ , how much is bought?" *and* "for quantity  $q$ , what's the most a buyer will pay?" Here the line hits  **$p = 10$  when  $q = 0$**  (nobody buys above €10) and slides down to more litres as price falls.

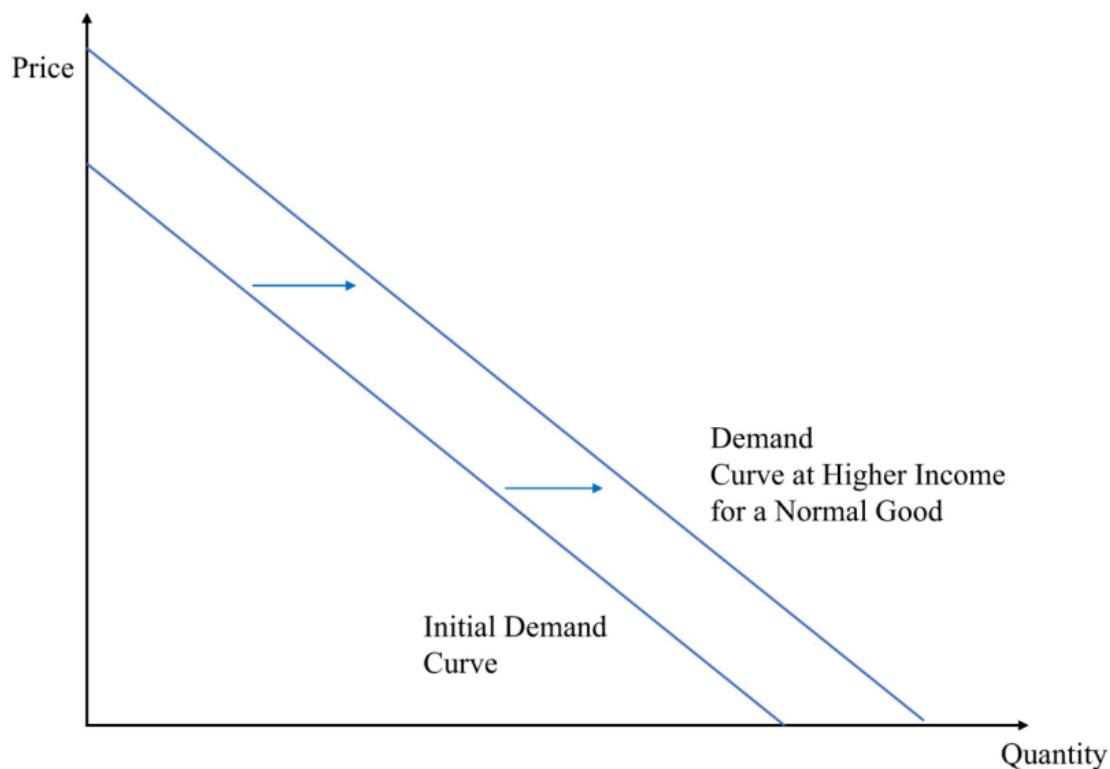
- ★ **Movement ALONG vs SHIFT — the make-or-break distinction:**
  - Own price changes → **MOVE ALONG** the existing curve (you slide to a new point).
  - Anything else changes → the whole curve **SHIFTS** (a *new* curve).



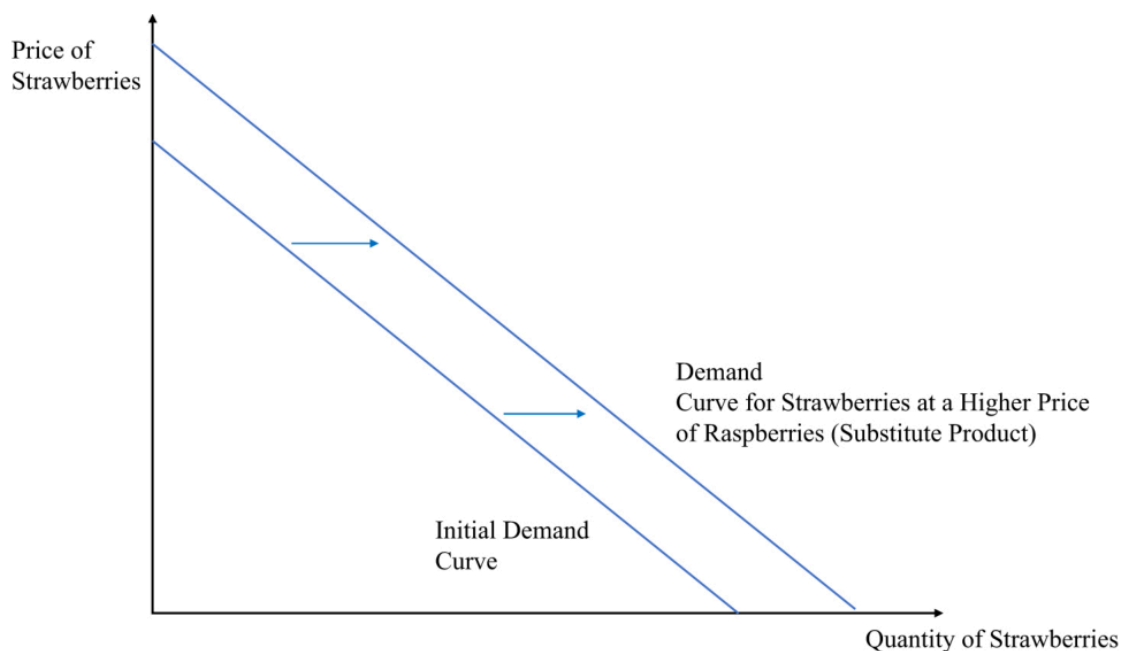
**How to read it (movement along).** Only the **own price** moved (7→6), so you **stay on the same curve** and just slide to a new point (q 3→4). No new curve is drawn. Contrast the shifts below, where the curve itself jumps.

- **Demand shifters** (each moves the *whole* curve = changes the intercept a):

| Shifter                                                                   | Direction                | Why                                           |
|---------------------------------------------------------------------------|--------------------------|-----------------------------------------------|
| <b>Income</b> ↑ (normal good)                                             | shift <b>out</b> (right) | richer → buy more at every price              |
| <b>Income</b> ↑ (inferior good, e.g. instant noodles)                     | shift <b>in</b> (left)   | richer → buy <i>less</i> of it                |
| <b>Price of a substitute</b> ↑ (raspberries ↑ → strawberry demand)        | shift <b>out</b>         | switch toward the now-relatively-cheaper good |
| <b>Price of a complement</b> ↑ (cereal ↑ → milk demand)                   | shift <b>in</b>          | the pair is consumed together                 |
| <b>Tastes / expectations</b> (heatwave → ice-cream; COVID → toilet paper) | out or in                | common-sense direction                        |



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**How to read them (shifts).** The whole line moves **right (out)** = more demanded at *every* price. Fig 2.4: higher **income** (normal good). Fig 2.6: a **substitute** (raspberries) got dearer, so buyers swing to strawberries. Note you are *not* sliding along the old curve — a brand-new curve sits to the right.

- **Substitutes vs complements (definitions):** **substitutes** — demand for good 1 **rises** when good 2's price rises (raspberries ↔ strawberries). **Complements** — demand for good 1 **falls** when good 2's price rises.

price rises (milk  $\leftrightarrow$  cereal).

- **From individual to aggregate demand (deck).** Market demand = **horizontal sum** of every buyer's demand — *add the quantities at each price*:

#### AGGREGATE (MARKET) DEMAND

$$Q(p) = q_1 + q_2 + \dots = D_1(p) + D_2(p) + \dots \text{ (add quantities across buyers at the same price)}$$

*Reads:* market demand = every individual's demand added up **at each price**. —  $Q(p)$  = total quantity the whole market wants at price  $p$  •  $q_1, q_2, \dots$  = the quantities buyer 1, buyer 2, ... each want at *that same* price •  $D_1(p), D_2(p)$  = each buyer's own demand function. (It's a **horizontal** sum — fix the price, add the quantities.)

*Deck example:* Mike's inverse demand  $p = 100 - 2q_M$ , Linda's  $p = 100 - \frac{1}{2}q_L \rightarrow$  invert each to  $q(p)$ , then add: at each price, total  $q = q_M + q_L$ . (Watch for a **kink** where one buyer drops out at high prices.)

- **The T-shirt trap (don't confuse products):** seeing pricey Gucci shirts outsell cheap generics is **not** an upward-sloping demand curve — those are **different goods**. A demand curve compares a good to **itself** (Gucci cutting *its own* price sells *more*). Real upward-sloping demand = the very rare **Giffen good** (Irish-famine potatoes; poor Chinese rice households — Jensen & Miller 2008).

## 2.2 Supply — the supply function

**Core:** supply answers "how much do firms bring to market at each price?" Firms are **price-takers** (they choose quantity, not the price). The curve usually slopes **up** — higher price  $\rightarrow$  more supplied.

- *In plain terms — why upward:* a higher price (i) makes it worth producing more even as costs rise, and (ii) draws in higher-cost producers who couldn't profit at the low price.
- **The supply function** (quantity supplied depends on price and input/other factors):

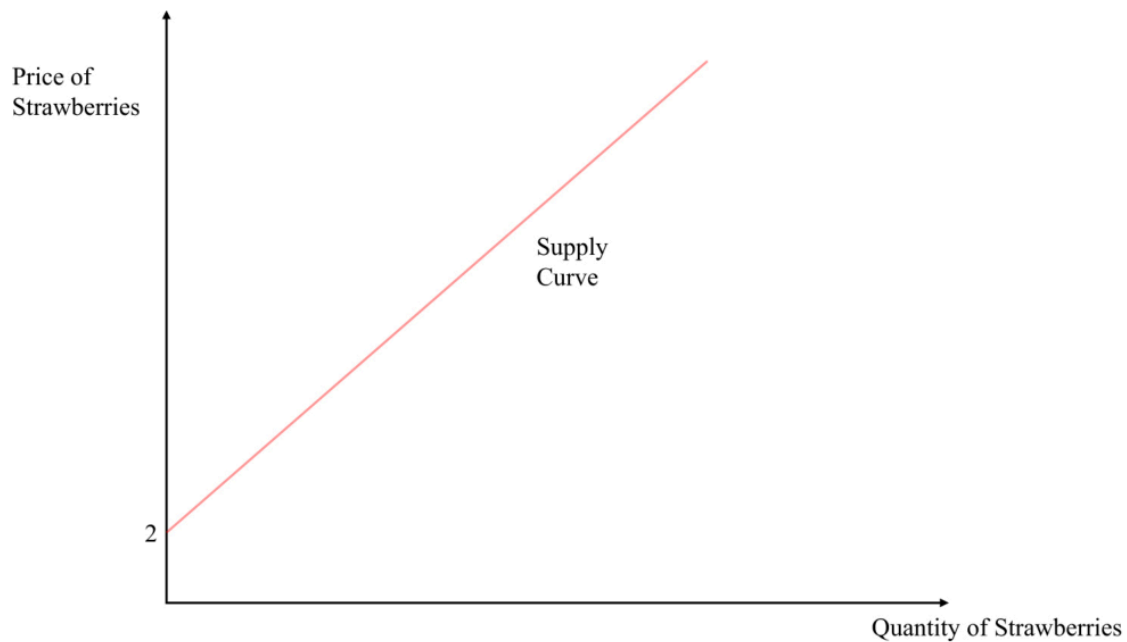
#### SUPPLY FUNCTION & ITS INVERSE

$$Q_S = S(p, p_o) \cdot \text{linear example: } Q_S = -2 + p \Rightarrow \text{inverse supply } p = 2 + Q_S$$

*Reads:* how much firms bring to market depends on the price they get and their costs. —  $Q_S$  = quantity supplied •  $S(\cdot)$  = "is a function of" •  $p$  = the good's **price** •  $p_o$  = price of **inputs** (wages, materials — higher costs shift supply in). In the example the "-2" is the price-intercept read backwards: **price must exceed 2 before any is supplied** (at  $p = 2, Q_S = 0$ ).

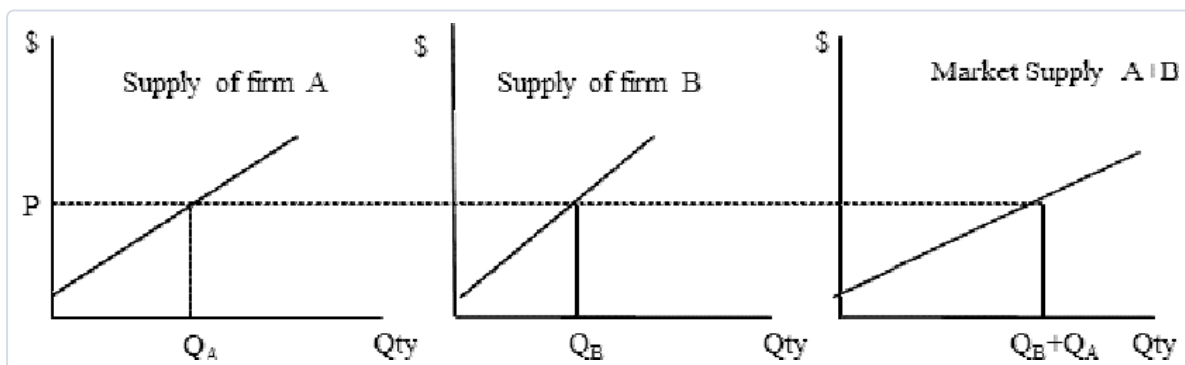
- **Why the "-2" (a negative intercept that isn't weird):** read it in **inverse** form — **price must clear €2 before any is supplied** (at  $p = 2, Q_S = 0$ ). *Check:*  $p = 3 \rightarrow Q_S = 1$ ;  $p = 4 \rightarrow Q_S = 2$ .

## Microeconomics



**How to read it (the supply curve).** Start at the €2 **price-intercept** (below it, zero supply) and climb: each extra euro of price brings one more litre to market. Like demand, an **own-price** change = **move along**; anything else = **shift**.

- **Supply shifters (move the whole curve):** input/cost  $\uparrow$  (wages, transport)  $\rightarrow$  shift **in** (need a higher price for the same quantity); **cheaper inputs / new tech / more sellers / market opens to foreign suppliers**  $\rightarrow$  shift **out**; **disruptions** (strikes, extreme weather, mine closures)  $\rightarrow$  shift **in**.
- **From individual to aggregate supply (deck):** market supply = **horizontal sum** of each firm's supply — add quantities across firms at each price.



**How to read it (aggregate supply).** At any price  $P$ , read each firm's quantity off its own curve ( $Q_A$ ,  $Q_B$ ) and **add them across** — the market curve is the sum, so it's **flatter** (more responsive) than any single firm's.

## ★ 2.3 Market equilibrium — solving for $p^*$ and $q^*$

**Core:** equilibrium = the **one price where quantity demanded = quantity supplied** ( $Q_D = Q_S$ ), the **market-clearing price** — no shortage, no glut, nobody wants to move. **This is the calculation to master.**

- The recipe (memorise these three steps):

STEP 1 — SET DEMAND = SUPPLY

$$Q_D = Q_S \Rightarrow 10 - p = -2 + p$$

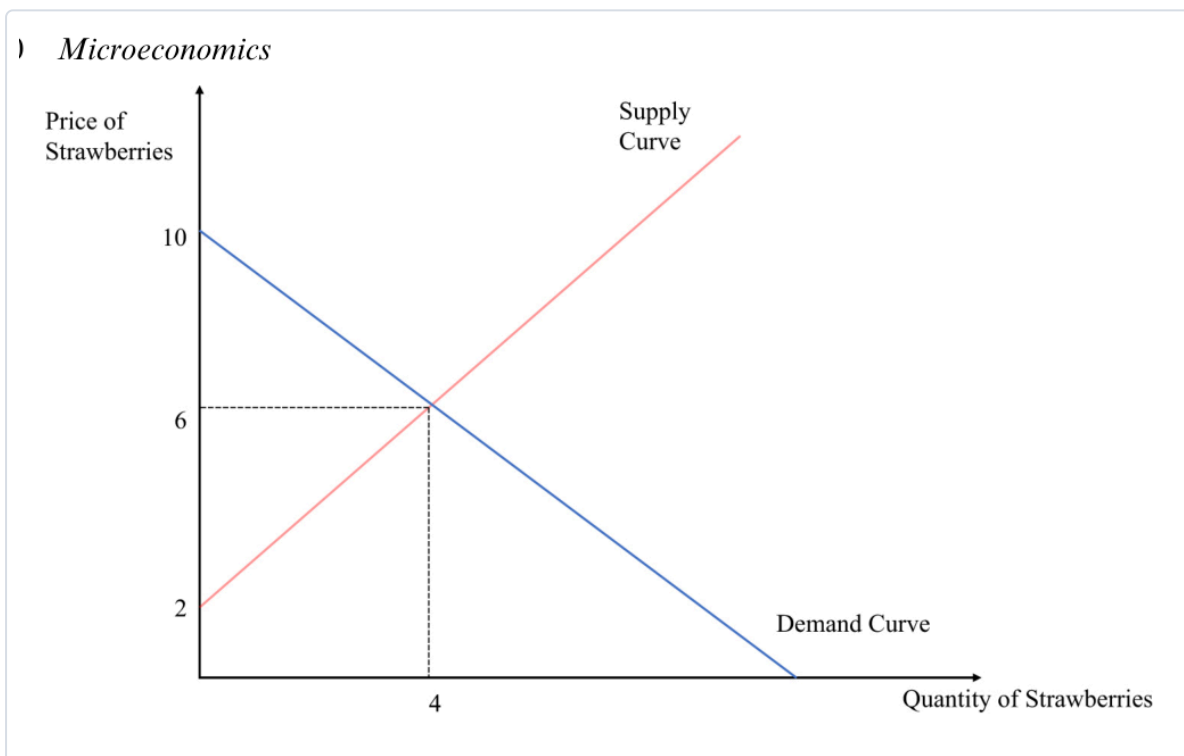
STEP 2 — SOLVE FOR THE EQUILIBRIUM PRICE  $P^*$

$$10 + 2 = p + p \Rightarrow 12 = 2p \Rightarrow p^* = 6$$

STEP 3 — PUT  $P^*$  BACK INTO EITHER CURVE FOR  $Q^*$

$$Q^* = 10 - 6 = 4 \text{ (check: } Q_S = -2 + 6 = 4 \checkmark) \Rightarrow \text{equilibrium } (p^*, q^*) = (6, 4)$$

*Reads:* find the price where the two plans agree, then the quantity that trades there. —  $Q_D$  = quantity demanded •  $Q_S$  = quantity supplied • setting  $Q_D = Q_S$  solves for  $p^*$  = the market-clearing **price**; substituting  $p^*$  back into either curve gives  $q^*$  = the **quantity** traded. The star (\*) just marks "equilibrium value." (Both curves must give the *same*  $q^*$  — that's the  $\checkmark$  check.)



**How to read it (equilibrium).** The **crossing point** is the only place the two plans agree — buyers want exactly what sellers offer (4 litres at €6). Everywhere else, one side is frustrated and price gets pushed toward the cross.

- **Off-equilibrium: excess demand & excess supply** (this is also *exactly* how you handle price controls below):

EXCESS DEMAND (SHORTAGE) – PRICE BELOW  $P^*$

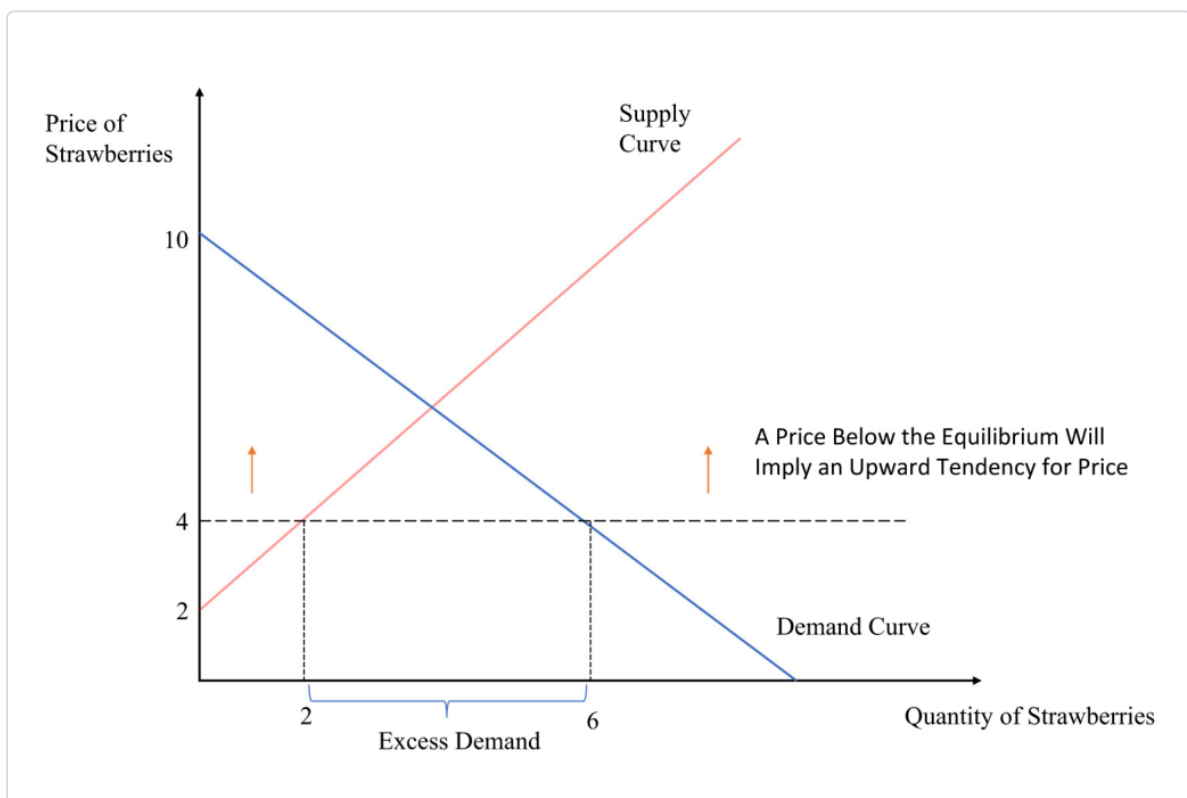
at  $p = 4$ :  $Q_D = 10 - 4 = 6$ ,  $Q_S = -2 + 4 = 2 \Rightarrow \text{shortage} = Q_D - Q_S = 4 \rightarrow$   
upward pressure on price

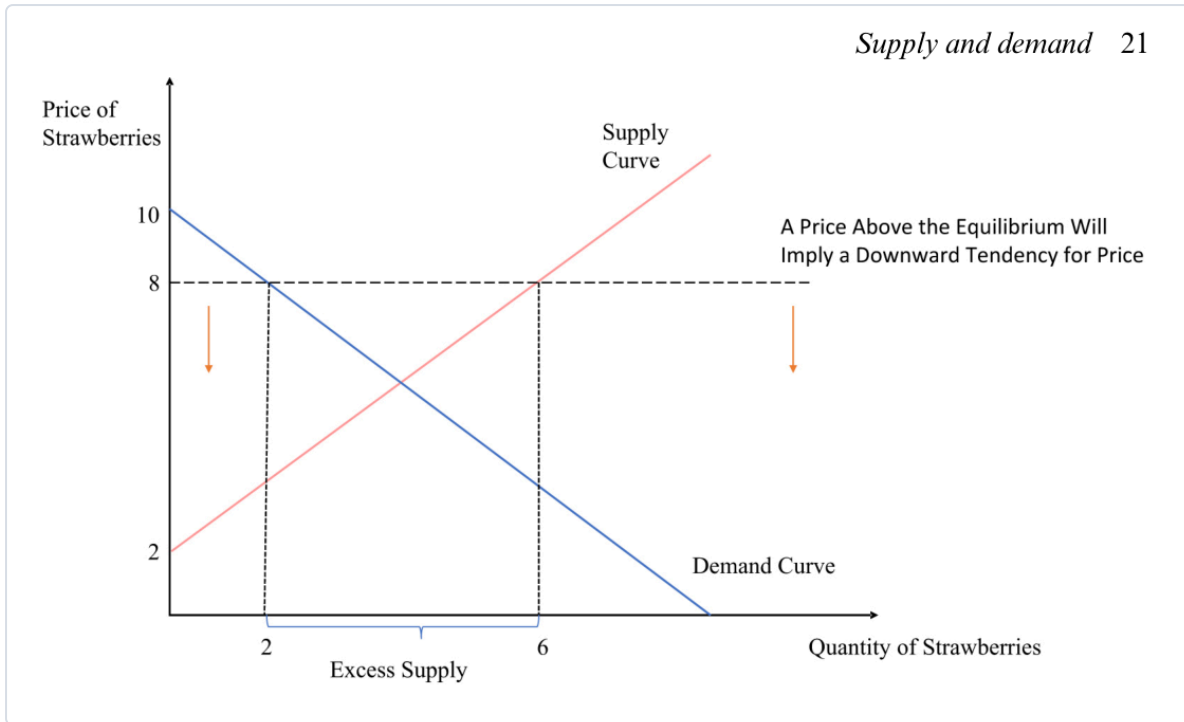
*Reads:* below equilibrium, buyers want more than sellers will supply. — plug the low price ( $p = 4$ ) into **both** curves  $\rightarrow Q_D = 6$  wanted,  $Q_S = 2$  offered; the gap  $Q_D - Q_S = 4$  is the **shortage**, which pushes the price **up** toward  $p^*$ .

EXCESS SUPPLY (SURPLUS) – PRICE ABOVE  $P^*$

at  $p = 8$ :  $Q_D = 10 - 8 = 2$ ,  $Q_S = -2 + 8 = 6 \Rightarrow \text{surplus} = Q_S - Q_D = 4 \rightarrow$   
downward pressure on price

*Reads:* above equilibrium, sellers offer more than buyers want. — plug the high price ( $p = 8$ ) into **both** curves  $\rightarrow Q_S = 6$  offered,  $Q_D = 2$  wanted; the gap  $Q_S - Q_D = 4$  is the **surplus**, which pushes the price **down** toward  $p^*$ .





**How to read them.** Pick the off-equilibrium price, drop a vertical line, and read **both** curves: the **horizontal gap** between them is the shortage (demand side wider) or surplus (supply side wider). The gap is what pushes price back toward  $p^*$ .

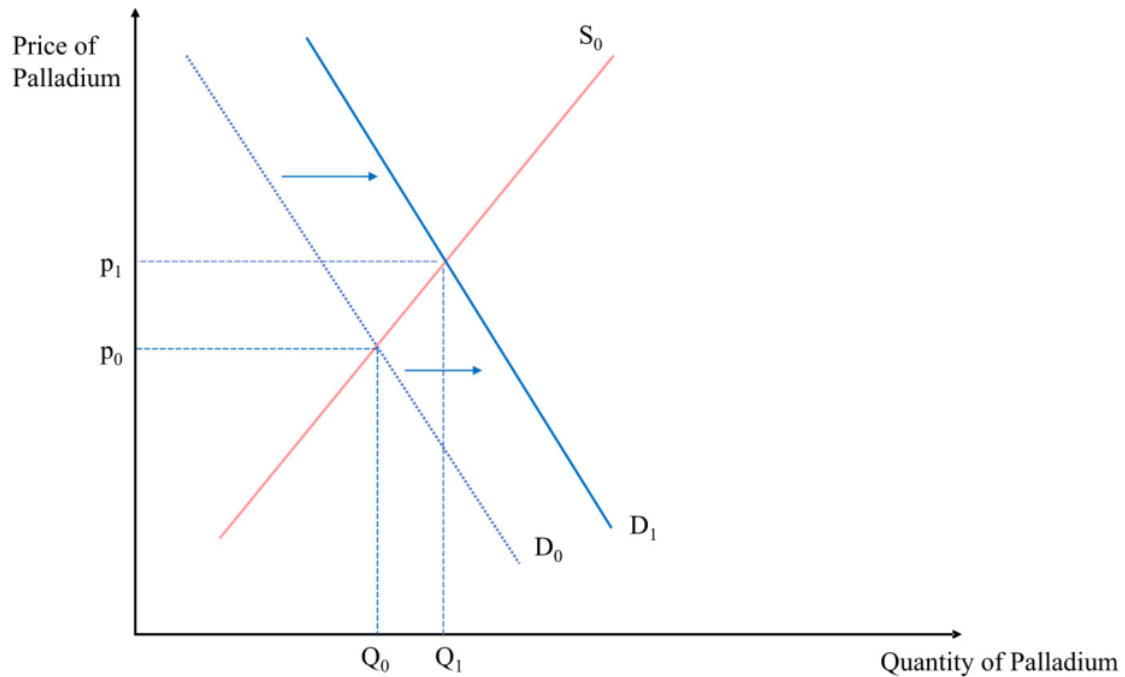
- **Marshall's scissors:** asking whether *demand or supply* sets the price is like asking *which blade of a scissors cuts* — **both do**. This dissolves the **diamond–water paradox**: water is cheap not because it's little wanted but because it's **abundant in supply**; diamonds are dear because they're **scarce**.
- (Who actually "sets" the price if everyone's a price-taker? The theoretical **Walrasian auctioneer** — and **Vernon Smith's 1962 experiments** showed real people reach the equilibrium price fast.)

## 2.4 Shifts in the equilibrium — comparative statics

**Core:** to analyse any shock, **don't jump to price/quantity**. First figure out **which curve moves and which way**; the effect on  $p^*$  and  $q^*$  follows.

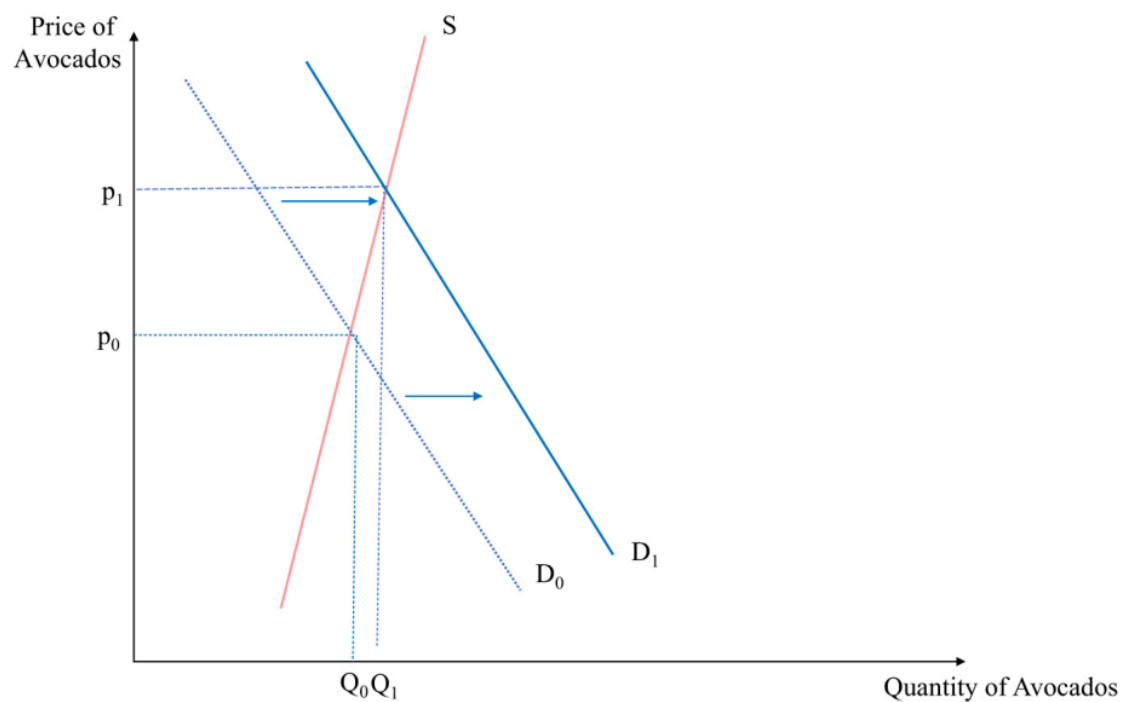
- **The method (Butera/Marshall — one change at a time, *ceteris paribus*):** ① Demand or supply? (falling incomes/other-good prices → demand; input costs/disruptions → supply). ② **Which direction** does it move *at a given price*? ③ *Then* read the new crossing. ④ **Reminder:** a *shift* is not a movement along — so a demand shift giving **both** higher price **and** higher quantity does **not** break the law of demand.

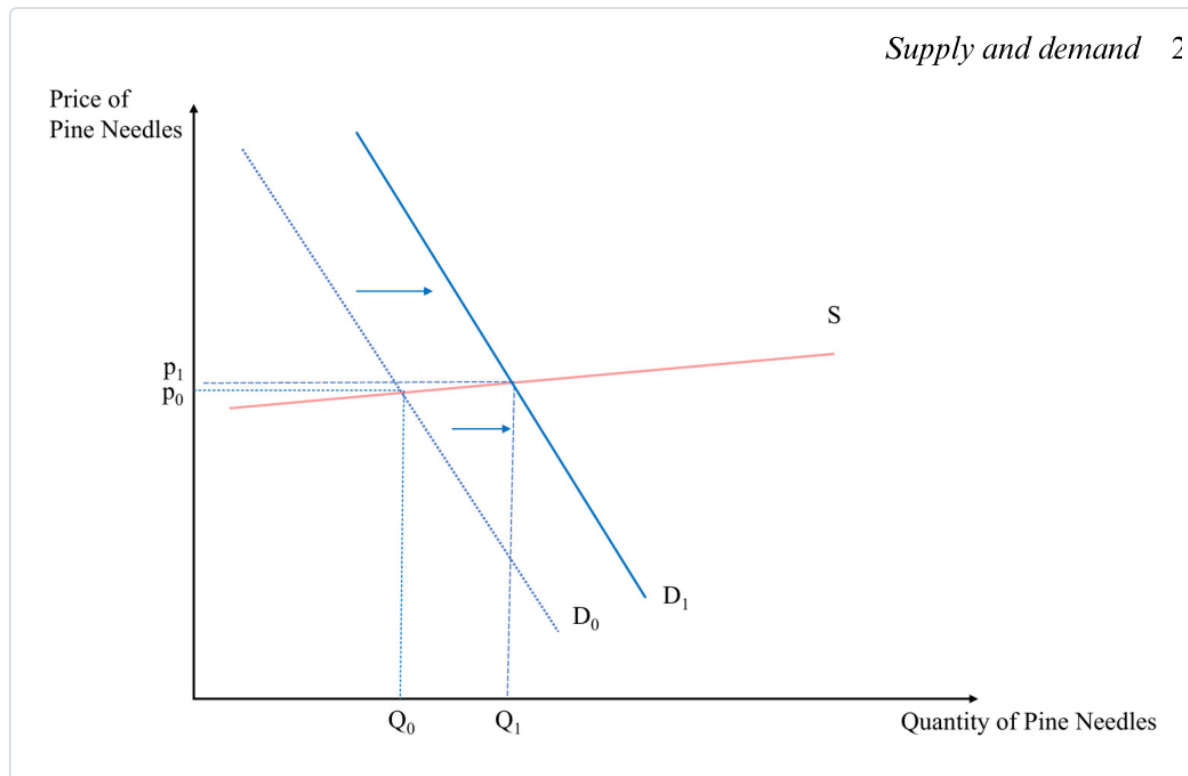
## Supply and demand



**How to read it (a demand shock).** Demand shifts **out** ( $D_0 \rightarrow D_1$ ); supply is unchanged, so producers **move along** supply to the new cross — **both  $p$  and  $q$  rise**. (Add a simultaneous supply **fall** and price rises *unambiguously*, but the **quantity effect becomes ambiguous** — it depends which shift is bigger.)

- ★ **Slope decides the *size* of the effect.** A demand shock's impact depends on the **slope of the supply curve**:





**How to read them.** Same outward demand shift, opposite results. **Steep supply** (avocados — trees take years, stock is fixed short-run): price jumps, quantity barely moves. **Flat supply** (pine needles — anyone can gather more): quantity jumps, price barely moves. *The flatter the other curve, the more a shock shows up as quantity; the steeper, the more it shows up as price.*

## Price ceilings & floors — when the government fixes the price (deck)

**Core:** a **binding price control** stops the market clearing, and you find the resulting gap with the **exact excess-demand/supply calculation above** — just set  $p$  to the controlled price.

- **Price ceiling** = a legal **maximum** (e.g. **rent control**). To bind it must be **below  $p^*$** , so  $Q_D > Q_S$  → a **shortage** (queues, waiting lists, black-market resale). *Compute:* plug the capped  $p$  into both curves; shortage =  $Q_D - Q_S$ .
- **Price floor** = a legal **minimum** (e.g. the **minimum wage** in the labour market). To bind it must be **above  $p^*$** , so  $Q_S > Q_D$  → a **surplus** (unsold goods; in the labour market, **unemployment** — more people want to work than firms will hire). *Compute:* surplus =  $Q_S - Q_D$ .
- **The trade-offs (positive, not normative): ceilings** — **pro:** keep essentials affordable for the less-well-off; **con:** shortages, quality decline, and markets "find a way around" (rent control can even be **regressive**; Venezuela's caps produced empty shelves). **floors** — **pro:** a living wage; **con:** firms substitute **capital for labour** (self-checkout at Føtex), cut hours, or make jobs temporary. "Good economics generates **positive** statements — evidence on costs and benefits — to inform the **normative** political choice."

## Is the model realistic? — perfect competition (deck)

**Core:** the supply-demand model assumes **perfect competition** — "every model is wrong, but useful."

- **Assumptions:** ① every agent is a **price-taker**; ② goods are **homogeneous** (perfect substitutes — one farm's wheat = another's; Shell petrol = Exxon petrol); ③ **perfect information** about prices/quality; ④ **low transaction costs**; (and many buyers & sellers). **Break these** and you need refinements — **monopoly, market failures** (later chapters).

## Cases & examples

Each: *the example* → *the concept it teaches*.

- **Strawberries (the running example)** → the whole toolkit:  $Q_D = 10 - p$ ,  $Q_S = -2 + p$ , equilibrium (6, 4), and the €4/€8 shortage/surplus.
- **Diamond–water paradox** → price is set by **supply and demand** (Marshall's scissors), not usefulness — water's cheap because it's abundant.
- **Palladium** (tighter emission rules → catalytic-converter demand ↑; S. African mine disruptions → supply ↓) → **comparative statics** with one and two curves moving.
- **2008 world rice spike & Haiti riots** (India's export ban cut supply; panic stockpiling shifted demand out) → a real **double shift**; prices later fell as stockpiles released + the financial crisis hit demand.
- **Instant noodles** → **inferior good**; **raspberries** → **substitute**; **cereal & milk** → **complements**.
- **Giffen goods** (Irish-famine potatoes; poor Chinese rice households) → the rare **upward-sloping demand** exception.
- **COVID toilet-paper stockpiling** → an **expectations** shift of demand.
- **Avocados vs Swedish pine-needle tea** → **steep vs flat supply** sizing a demand shock.
- **Rent control / Venezuela** → **price ceiling** → **shortage**; **minimum wage** → **price floor** → **surplus** (unemployment).

## Formula sheet — quick reference

| Object                    | Formula                                                                  | Note                                     |
|---------------------------|--------------------------------------------------------------------------|------------------------------------------|
| Demand function           | $Q_D = D(p, p_s, p_c, Y, \tau)$                                          | many variables; curve fixes all but p    |
| Linear demand / inverse   | $Q_D = a - b \cdot p / p = (a - Q_D)/b$                                  | a = intercept (shifters), b = slope      |
| Supply function / inverse | $Q_S = -2 + p / p = 2 + Q_S$                                             | negative intercept = min price to supply |
| Aggregate (market)        | $Q(p) = \sum D_i(p) ; \sum S_i(p)$                                       | <b>horizontal</b> sum (add quantities)   |
| <b>Equilibrium</b>        | <b>set <math>Q_D = Q_S \rightarrow p^*</math>, then <math>q^*</math></b> | the market-clearing point                |
| Excess demand (shortage)  | $Q_D - Q_S$ at $p < p^*$                                                 | → upward price pressure; price ceiling   |
| Excess supply (surplus)   | $Q_S - Q_D$ at $p > p^*$                                                 | → downward pressure; price floor         |

## Exam pointers

- **Solve a market from scratch:** given a linear demand and supply, **set them equal**, get  $p^*$ , back-substitute for  $q^*$  — the strawberry case ( $p^*=6$ ,  $q^*=4$ ) is the model answer.

- Compute a **shortage/surplus**: plug a below- or above-equilibrium price into *both* curves and take the gap ( $Q_D - Q_S$  or  $Q_S - Q_D$ ) — and know a **ceiling**→shortage, **floor**→surplus.
- Diagnose movement vs shift, and for a shock say **which curve moves, which way**, and the effect on  $p^*$  &  $q^*$  (incl. the **ambiguous-quantity** double-shift and the **slope** point).
- Explain **why demand slopes down** (scarcity + diminishing marginal utility) and **Marshall's scissors** / the diamond–water paradox.
- Name the **perfect-competition** assumptions and why the model is still "useful though wrong."

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## ★ Fun facts & memorable details (Lecture 2)

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Sticky bits from Supply & Demand.

- **Marshall's scissors**: asking whether demand or supply sets the price is like asking **which blade of the scissors** does the cutting — both do.
- The **diamond–water paradox** is only a paradox if you forget supply: water is life-or-death yet cheap **because it's abundant**.
- **Giffen goods** really exist: when their staple got dearer, poor Chinese rice households bought **more** rice, not less (Jensen & Miller 2008) — they could no longer afford anything else.
- A negative supply intercept ( $Q_S = -2 + p$ ) isn't a mistake — it just means **price must top €2 before anyone supplies a drop**.
- **Vernon Smith** won a Nobel (2002) for showing that real people in a lab **converge on the equilibrium price** startlingly fast — the "Walrasian auctioneer" made flesh.
- Rent control's dirty secret: it can be **regressive** and spawn black-market "key money" — a "tale of good intentions" (cf. Venezuela's empty shelves).
- Minimum wage as a price floor nudges firms toward **capital over labour** — the **self-checkout at Føtex** is a supply-and-demand diagram in the wild.

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## Lecture 3: Elasticities & Taxes — the Supply & Demand model II (Ch. 2 recap · §6.4)

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**Required reading:** Friberg, §6.4 Elasticities (own-price, cross-price, income; supply elasticity) — building on the **Chapter 2** supply-and-demand model (see [Lecture 2](#)). *The lecture deck also teaches taxes on goods & services — that part draws on the textbook's §7.5 (unit tax, tax incidence), covered here because the deck foregrounds it.*

**Theme of the lecture** (Butera — from the deck, titled "*Supply and Demand model*"). Chapter 2 taught us to **move the curves around**; this lecture adds **precision**. The *"shape"* of a demand or supply curve — its elasticity — *decides how much a shock (or a tax) shows up as a change in price versus a change in quantity*. The deck's

three goals: (1) understand what demand & supply **elasticities** mean, (2) **derive** them from demand/supply curves, and (3) use them to analyse the **effects of taxes**.

★ **The heart of the lecture: elasticity = % responsiveness** — "by what % does quantity change when price rises 1%." It's a **unit-free** number that summarises a curve's **shape**, so it's comparable across markets and time. The **shape decides how a shock splits into price vs quantity**: with **inelastic** demand a shock/tax hits **price**; with **elastic** demand it hits **quantity**. And a **tax** drives a **wedge** between the price buyers pay and sellers receive — **who bears it depends on the relative elasticities**, and (in perfect competition) **it doesn't matter who is legally taxed**.

**The red thread.** For any market ask: *how responsive is quantity to price (elastic or inelastic)?* — because that single question answers **how a supply shock moves the market**, **how a firm should price**, and **who really pays a tax**. Elasticity is the "shape" number that ties all three together.

## Key concepts / "modes" to use

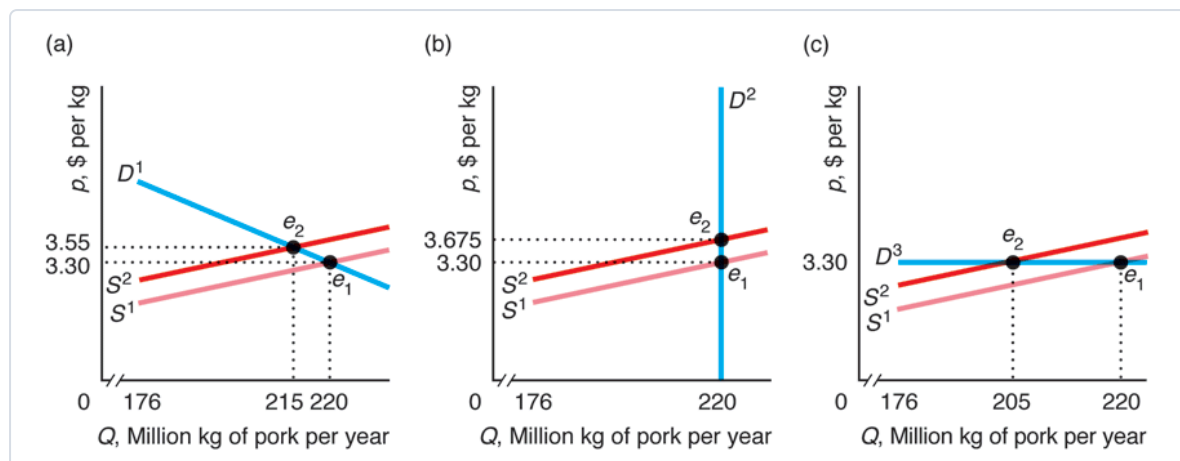
- **Elasticity** — the % change in one variable ÷ % change in another (a unit-free number).
- **Price elasticity of demand ( $E_D$ )** — % change in quantity demanded ÷ % change in own price (negative).
- **Elastic / inelastic / unit-elastic** —  $|E| > 1$  /  $|E| < 1$  /  $|E| = 1$ .
- **Point-elasticity trick** —  $E_D = (dQ/dp) \cdot (p/Q)$  ; on a linear curve elasticity **changes along the curve** (constant-elastic curves don't).
- **Cross-price elasticity** — substitutes (+) vs complements (-).
- **Income elasticity** — normal (+) vs inferior (-); **Engel curves / Engel's law**.
- **Supply elasticity** — % change in quantity supplied ÷ % change in price (positive).
- **Tax wedge & tax incidence** —  $p_b = p_s + t$  ; who bears the burden depends on **relative elasticities**; legal  $\neq$  economic incidence; unit vs ad-valorem tax; **deadweight loss**.

## Section-by-section main points

### Why do we care about elasticity? (deck)

**Core:** the **shape** of demand/supply curves determines **how much a shift in one moves the equilibrium** — and "shape" is summarised by **elasticity**. A steep (inelastic) curve and a flat (elastic) curve respond to the *same* shock very differently.

- **The motivating questions (deck):** Do "**sin taxes**" (cigarettes, sugar) hurt the poor? Why do firms **raise prices** even when it looks crazy (Netflix, Amazon Prime)? To what extent are **taxes on producers "passed on"** to consumers? Every answer runs through elasticity.
- **The visual intuition:** how far a **supply shift** moves price vs quantity depends entirely on the **slope/elasticity of demand** — see the figure below (perfectly inelastic demand → only price moves; perfectly elastic demand → only quantity moves).



**How to read it (why elasticity matters).** All three panels apply the **same supply shift**  $S^1 \rightarrow S^2$ . The **only thing** that differs is the **demand curve's shape**:

- **(a) Ordinary downward-sloping demand:** the shock splits into **both** a higher price *and* a lower quantity.
- **(b) Perfectly inelastic demand (vertical):** quantity can't respond, so the shock lands **entirely on price** ( $3.30 \rightarrow 3.675$ ). *This is the "necessity with no substitutes" case.*
- **(c) Perfectly elastic demand (horizontal):** price can't move, so the shock lands **entirely on quantity** ( $220 \rightarrow 205$ ).
- **Take-away:** the **more inelastic** the demand, the more a shock (or tax) shows up as **price**; the **more elastic**, the more it shows up as **quantity**. *This one picture is the whole reason we measure elasticity.*

## ★ Price elasticity of demand — definition

**Core:** the **price elasticity of demand** measures how sensitive quantity demanded is to the good's own price — the **% change in quantity**  $\div$  **% change in price**. It is a **pure (unit-free) number** and is **negative** (demand slopes down: price up  $\rightarrow$  quantity down).

### PRICE ELASTICITY OF DEMAND

$$E_D = ( \Delta Q / Q \times 100 ) \div ( \Delta p / p \times 100 ) = \% \Delta Q \div \% \Delta p$$

**Reads:** how responsive quantity demanded is to a price change — "a 1% rise in price  $p$  leads to an  $E_D$  % change in quantity  $q$ ." —  $\Delta Q / Q \times 100$  = the **percentage change in quantity demanded** ·  $\Delta p / p \times 100$  = the **percentage change in price** ·  $E_D$  = the ratio (a **unit-free number**, **negative** for a normal good). The  $\times 100$ s cancel, so it's just **% $\Delta Q$  over % $\Delta p$** .

- **Why unit-free matters (the point of the whole concept):** raw slopes aren't comparable — "1,000 tons less rice" means something different in **Vietnam** vs **Bhutan**, and depends on the currency. A **percentage measure** is **comparable across markets, currencies and time**.
- **Why negative:** demand curves slope **down**, so  $\Delta Q$  and  $\Delta p$  have opposite signs  $\rightarrow$  the ratio is negative. (When people say a "higher" elasticity they usually mean a bigger **absolute value**  $|E_D|$ .)
- **Worked example:** a café raises its coffee price **10%** and sells **8% fewer** cups  $\rightarrow E_D = -8\% \div +10\% = -0.8$ . Since  $|-0.8| < 1$ , demand is **inelastic** — revenue actually **rises** (price up 10%, quantity down only 8%). (Full step-by-step computations are in the next section.)

## Computing elasticity — and the point-elasticity trick

**Core:** plug the two percentage changes into the formula. On a linear demand curve the elasticity is *not* constant — it depends on *where* you evaluate it, via  $E_D = (dQ/dp) \cdot (p/Q)$ .

- **Simple example (textbook Table 6.2):** quantity falls  $6 \rightarrow 4$  while price rises  $1 \rightarrow 2$ . Then

### WORKED — SIMPLE EXAMPLE

$$E_D = ((4-6)/6) \div ((2-1)/1) = (-2/6) \div (1) = -1/3$$

*Reads:* a demand that is **inelastic** —  $(4-6)/6 = -0.33$  = the % fall in quantity ( $-33\%$ )  $\cdot$   $(2-1)/1 = 1$  = the % rise in price ( $+100\%$ )  $\cdot -1/3$  = quantity falls **only**  $\frac{1}{3}$  as fast as price rises  $\rightarrow$  **inelastic** demand.

- **A policy-style example (soda, Allcott et al. 2019):** own-price elasticity of sugary soda  $\approx -1.37$ . If price rises 5%, quantity changes by  $E_D \times 5\% = -1.37 \times 5 = -6.85\%$  (a  $\sim 7\%$  fall). To cut demand 15%, price must rise  $-15 \div -1.37 \approx 11\%$ .  $\rightarrow$  this is exactly the sum a policymaker designing a **soda tax** needs.
- **The point-elasticity trick (linear demand):** rewrite the formula as

### POINT ELASTICITY

$$E_D = (\Delta Q / \Delta p) \cdot (p/Q) = (dQ/dp) \cdot (p/Q)$$

*Reads:* elasticity = **slope of the demand function**  $\times$  the **price-to-quantity ratio at your chosen point** —  $dQ/dp$  = the **slope** (how many units Q changes per €1 of price; constant for a straight line)  $\cdot$   $p/Q$  = the **price-to-quantity ratio** where you evaluate it (this is what varies along the curve).

- **Worked ( $Q = 10 - 2p$ ):** here  $dQ/dp = -2$  (always). But the **elasticity varies**:
  - at  $p = 2.5$  (so  $Q = 5$ ):  $E_D = -2 \times 2.5/5 = -1$  (**unit elastic**).
  - at  $p = 4$  (so  $Q = 2$ ):  $E_D = -2 \times (p/Q) = -2 \times 4/2 = -4 \rightarrow$  **elastic**. ( $\triangle$  The textbook misprints this as  $-2 \times 2/2 = -2$  — it accidentally puts  $Q=2$  in the numerator instead of  $p=4$ ; the correct  $p/Q$  is  $4/2 = 2$ , so  $E_D = -4$ .)
  - **The pattern: demand is more elastic at higher prices** — at  $p=2.5$  it's unit-elastic ( $-1$ ), at  $p=4$  it's elastic ( $-4$ ). A fixed unit change is a bigger *percentage* change when Q is small and p is large.
- **Constant-elasticity demand (contrast):** the form  $Q = a \cdot p^\epsilon$  has the **same** elasticity  $\epsilon$  at *every* price (used in empirical estimation via  $\ln Q = \ln a + \epsilon \cdot \ln p + \text{error}$ ). So: **linear  $\rightarrow$  elasticity changes along the curve; constant-elastic  $\rightarrow$  it doesn't**.
  - **Worked example:** with  $Q = 100 \cdot p^{-1.2}$ , the elasticity is  $-1.2$  whether you evaluate at  $p = €1$  ( $Q = 100$ ),  $p = €5$  ( $Q \approx 15$ ), or  $p = €50$  — it never changes. (Contrast the linear  $Q = 10 - 2p$  above, where it swung from  $-1$  to  $-4$ .)

## Elastic, inelastic, unit-elastic — and why it matters

**Core:** we name the ranges of elasticity, and each has a sharp economic meaning for how supply shocks split into price vs quantity.

| Name         | Level (of E_D)               | Meaning                     | Curve looks...          |
|--------------|------------------------------|-----------------------------|-------------------------|
| Inelastic    | $-1 < E_D < 0$ ( $ E  < 1$ ) | %ΔQ <b>smaller</b> than %Δp | relatively <b>steep</b> |
| Unit elastic | $E_D = -1$                   | %ΔQ <b>equals</b> %Δp       | —                       |
| Elastic      | $E_D < -1$ ( $ E  > 1$ )     | %ΔQ <b>bigger</b> than %Δp  | relatively <b>flat</b>  |

- **Why it matters (the punchline):** with **inelastic** demand, **supply shocks translate into big price changes** (little quantity response) — critical for e.g. **cocoa** growers in Ghana/Côte d'Ivoire (world cocoa elasticity  $\approx -0.19$  to  $-0.96$ ; policy point estimate  $-0.34$ ). With **elastic** demand, shocks translate into **quantity** changes.
- **What makes demand elastic — substitutes.** The single biggest driver is **availability of close substitutes**: pharmaceuticals with few substitutes are **inelastic** (people pay whatever for a needed medicine); goods with many substitutes are **elastic**.
- **Aggregation level matters:** a *specific brand* is more **elastic** than the *category* (more substitutes for one brand). US beer: **product-level** median elasticity  $\approx -4.74$ , but beer **overall**  $\approx -0.60$ .

## Cross-price elasticity — substitutes vs complements

**Core:** the **cross-price elasticity** measures how demand for good **A** responds to a change in the price of a **different** good **B**. Its **sign** tells you the *relationship* between the two goods; its **size** tells you *how strong* that relationship is.

### CROSS-PRICE ELASTICITY

$$E_{cross} = \% \Delta Q_A \div \% \Delta p_B \rightarrow > 0 = \text{substitutes} \cdot < 0 = \text{complements} \cdot \approx 0 = \text{unrelated}$$

**Reads:** how the quantity of **A** responds to a price change in **B** — %ΔQ\_A = percentage change in demand for good A · %Δp\_B = percentage change in the price of good B · E\_cross = the ratio (can be + or –, unlike own-price which is always –).

- **The sign = the relationship:**
  - **Positive → substitutes.** B gets dearer → buyers switch to A → A's demand **rises** (butter ↑ → margarine demand ↑; Heineken ↑ → Carlsberg ↑).
  - **Negative → complements.** B gets dearer → buyers use less B *and* less of its partner A → A's demand **falls** (petrol ↑ → big-car demand ↓; smartphones ↑ → phone-case demand ↓).
  - $\approx 0 \rightarrow$  **unrelated** goods (the price of bread barely touches demand for haircuts).
- **The size = the strength:** the larger the **absolute value**  $|E_{cross}|$ , the **closer** the substitutes (or the stronger the complementarity). Two brands of the *same* soda have a **big** positive cross-price elasticity; soda and bottled water a **small** one; soda and umbrellas  $\approx 0$ .
- **Worked example:** Danish spirits get **10% dearer** → Swedish demand for spirits changes by  $E_{cross} \times 10\% = +0.3 \times 10 = +3\%$  (Swedes buy **more at home** instead of driving to Denmark) → **positive** → **substitutes** (Asplund 2007). *Flip it: if smartphones get 10% dearer and phone-case demand falls 6%,*  $E_{cross} = -6 \div 10 = -0.6 \rightarrow$  **negative** → **complements**.

- ★ **Its killer application — defining markets & judging mergers.** This is *the* reason cross-price elasticity matters in practice. Competition authorities use it to decide **whether two products are in the same market**: a **high positive** cross-price elasticity means A and B are **close substitutes**, so a firm selling both has real market power (a merger between them removes a competitor). This is the logic of the **SSNIP / "hypothetical monopolist" test** — *would a Small but Significant Non-transitory Increase in Price (say 5%) be profitable, or would enough buyers flee to substitutes to make it a loss?* If they flee, the substitute is in the same market.
- **Asymmetry (watch out):** cross-price elasticity is **not symmetric** — A's response to B's price  $\neq$  B's response to A's price. A cheap **complement** (phone case) reacts strongly to the price of the expensive main good (phone), but the phone barely reacts to the case's price.
- **Real numbers (Asplund et al. 2007):** Swedish **spirits +0.3, beer +0.5, wine +0.2** with respect to their **Danish** prices — all **positive**, i.e. all **substitutes** for cross-border shopping (beer the closest). (*Deck aside: cigarettes & alcohol  $\approx -1$  — strong complements, Krauss et al. 2014.*)

## Income elasticity — normal vs inferior goods, Engel curves

**Core:** the **income elasticity** measures how demand responds to **income** — its **sign** distinguishes **normal** from **inferior** goods.

### INCOME ELASTICITY

$$E_{income} = \% \Delta Q \div \% \Delta I \rightarrow > 0 = \text{normal good} \cdot < 0 = \text{inferior good}$$

*Reads:* how quantity demanded responds to an income change —  $\% \Delta Q$  = percentage change in quantity ·  $\% \Delta I$  = percentage change in income · **sign: positive  $\rightarrow$  normal** (demand rises with income), **negative  $\rightarrow$  inferior** (demand falls as income rises — e.g. instant noodles, store-brand goods).

- **Worked example:** a household's **income rises 10%** and its spirits purchases rise **14%**  $\rightarrow E_{income} = +14 \div +10 = +1.4 \rightarrow$  **normal and income-elastic** (a "luxury,"  $> 1$ ) (Asplund 2007). *Flip it: income +10% but instant-noodle demand -5%  $\rightarrow E_{income} = -5 \div 10 = -0.5 \rightarrow$  inferior.*
- **Nuance:** a good can be **normal at low incomes and inferior at higher ones** (hostels for a student: more nights as income rises, then switch to hotels). Tastes vary by person.
- **Uses:** long-term **projections** (healthcare, appliances in low-income countries) and predicting the **business cycle** (pharma is income-insensitive  $\rightarrow$  "defensive stock").
- **Engel curves / Engel's law (Ernst Engel, 1800s Prussian statistician):** plot demand (vertical) against income (horizontal) — **positive slope for normal goods, negative for inferior**. **Engel's law:** the **share of income spent on food falls** as income rises  $\rightarrow$  the income elasticity of food is **less than 1**.

## Supply elasticity — same principle

**Core:** the **price elasticity of supply** measures how sensitive quantity *supplied* is to price — same formula, but **positive** (supply slopes up).

## PRICE ELASTICITY OF SUPPLY

$$E_S = \% \Delta Q_S \div \% \Delta p \quad (\text{positive})$$

*Reads:* how responsive quantity supplied is to a price change —  $\% \Delta Q_S$  = percentage change in quantity supplied ·  $\% \Delta p$  = percentage change in price ·  $E_S$  = the ratio (**positive**: higher price → more supplied). A **steep** supply curve is **inelastic** (hard to expand output quickly); a **flat** one is **elastic**.

- **Worked example:** the market price of new flats rises **20%**, but the quantity of new flats supplied rises only **4%** this year →  $E_S = +4 \div +20 = +0.2$  → **inelastic** supply (you can't build quickly). Give it several years and builders respond more → supply becomes **more elastic**.

## Elasticity over time — short run vs long run

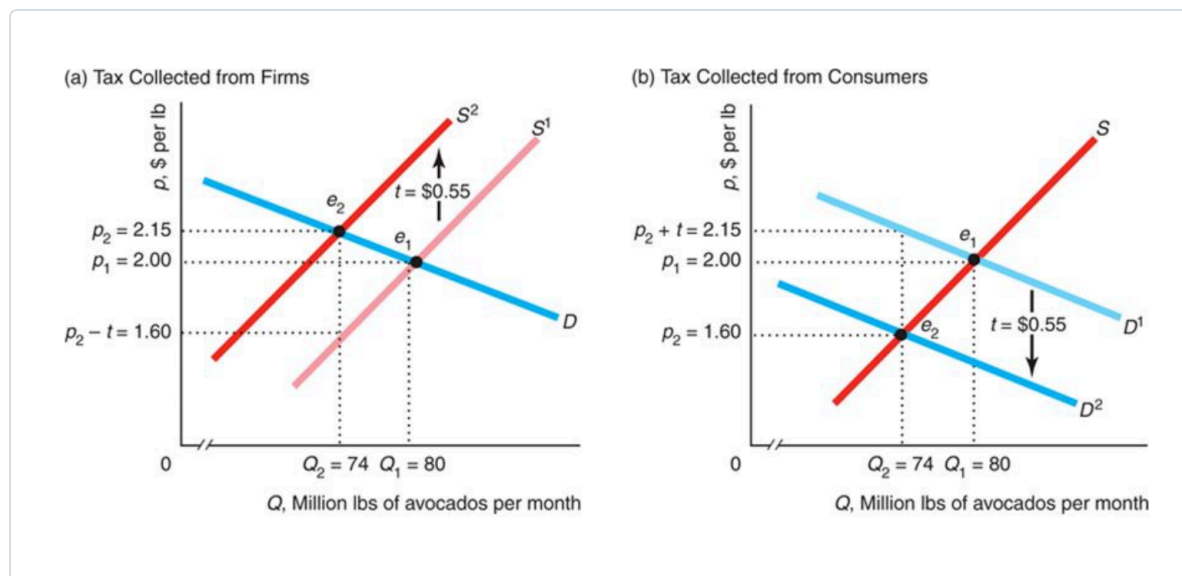
**Core:** demand (and supply) are usually **more elastic in the long run** than the short run, because people have **more time to adjust / find substitutes**.

- **Electricity (Labandeira et al. 2017):**  $\approx -0.24$  short-run vs  $-0.6$  long-run — over time consumers buy more efficient equipment, change habits.
- **The deck's cases:** a drop in **gas** prices does little in the short run but more in the long run; likewise **computers**. *Exam framing:* "in the short run demand is inelastic (people are locked in); given time they substitute, so long-run demand is more elastic."

## ★ Taxes on goods & services — the wedge, incidence & equivalence (deck; textbook §7.5)

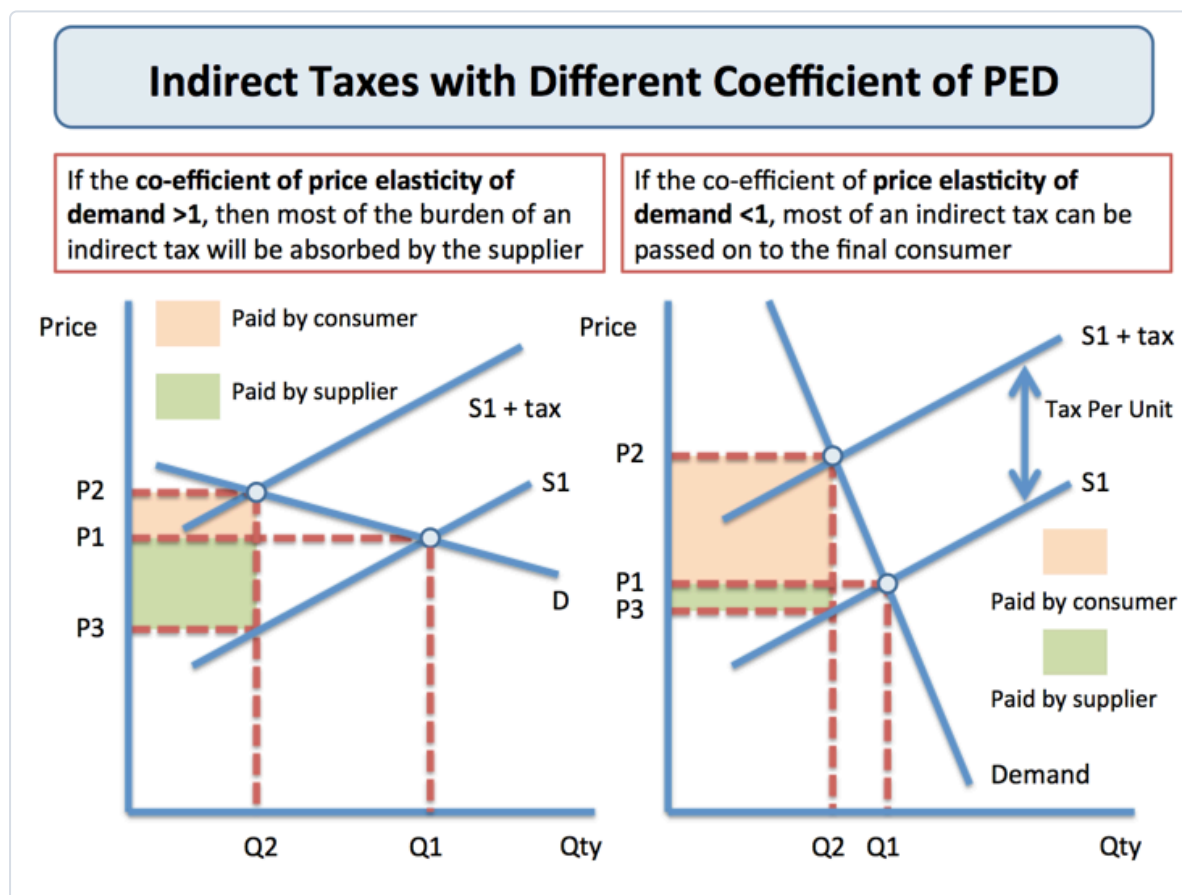
**Core:** a tax **drives a wedge** between the price **buyers pay** ( $p_b$ ) and the price **sellers receive** ( $p_s$ ):  $p_b = p_s + t$ . It raises the buyer price, lowers the seller price, cuts the quantity traded, and creates a **deadweight loss**. The split of the burden — **tax incidence** — depends on the **relative elasticities**, and (under perfect competition) **not** on who is legally taxed.

- **Two types of tax (deck):**
  - **Unit (specific) tax** — a fixed **amount  $t$  per unit** (per litre of petrol, per cigarette).
  - **Ad valorem tax** ("sales tax"/VAT) — a **fraction  $\alpha$  of the price/spend** (government keeps a share of each euro spent).
- **The wedge (worked unit-tax example,  $t = \text{€}3$ ):** with  $Q_D = 23 - p_b$ ,  $Q_S = -1 + p_s/2$ , and  $p_b = p_s + 3$ , solving gives  $p_s = 14$ ,  $p_b = 17$ ,  $Q = 6$ . Buyers pay **€3 more** than sellers receive; quantity falls from the no-tax equilibrium.
  - **Tax revenue** =  $t \times Q = 3 \times 6 = \text{€}18$ . The rest of the surplus lost is **deadweight loss** (here  $3 \times (7-6)/2 = \text{€}1.5$ ) — trades that *would* have happened but now don't.
- **Legal incidence  $\neq$  economic incidence (the key result).** It **doesn't matter** whether the tax is collected from **firms** (supply shifts up by  $t$ ) or **consumers** (demand shifts down by  $t$ ) — the quantity, the prices actually paid/received, and the DWL are **identical**. *All that matters is that the tax drives a wedge.* (Perfect-competition assumptions make this exact: no transaction costs, perfect information.)



**How to read it (tax equivalence).** Both panels impose the **same \$0.55 tax** on avocados. (a) collects it from **firms** → the supply curve shifts **up** by  $t$ ; (b) collects it from **consumers** → the demand curve shifts **down** by  $t$ . In **both** cases the result is identical: **Q falls to 74**, **buyers pay 2.15**, **sellers receive 1.60**, and the wedge between them is exactly **\$0.55**. → *The economics doesn't care who writes the cheque.*

- **What does decide the split: relative elasticities.** The more **inelastic** side of the market bears **more** of the tax (it can't escape by changing quantity). The more **price-sensitive (elastic)** side is affected less on price, **more** on quantity.



**How to read it (incidence & elasticity).** The tax raises the supply curve to  $S + \text{tax}$ . The **new higher price  $P_2$**  is what **consumers** pay;  **$P_3$**  is what **suppliers** keep. The split of the rectangle between them is

the **incidence**:

- **Left — elastic demand ( $PED > 1$ ):** consumers easily cut back, so firms **can't raise the price much** → the **supplier absorbs most** of the tax.
- **Right — inelastic demand ( $PED < 1$ ):** consumers keep buying regardless, so the firm **passes most of the tax through** → the **consumer bears most** of it.
- **Policy take-aways:** (1) "**sin taxes**" on inelastic goods (cigarettes, alcohol) fall mostly on **consumers** and raise a lot of revenue with **little DWL** (quantity barely moves) — which is *why* governments like them, and why they can be **regressive** (hit the poor). (2) We **prefer taxes on inelastic bases** — they distort choices least.
- **Subsidies are the mirror image:**  $p_s = p_b + s$  — a subsidy lowers the buyer price, raises the seller price, **expands** quantity above the efficient level, and also creates a **deadweight loss** (units produced that cost more than consumers value them). Its incidence, too, depends on elasticities (inelastic demand — e.g. emergency care, primary education — → small DWL).
  - **Worked example (subsidy  $s = €3$ ):** same market (  $Q_D = 23 - p_b$  ,  $Q_S = -1 + p_s/2$  ) but now  $p_s = p_b + 3$  → solving gives  $p_b = 15$ ,  $p_s = 18$ ,  $Q = 8$  — buyers pay **less** and sellers get **more**, and quantity *rises* above the no-subsidy level ( $6 \rightarrow 8$ ). The subsidy costs the government  $s \times Q = 3 \times 8 = €24$  , part of which is **deadweight loss**.

## Cases & examples

*Read each as a worked use of an elasticity number (or a tax result).*

- **Sugary-soda tax (Allcott et al. 2019).** Own-price elasticity  $\approx -1.37$  (elastic). A 5% price rise →  $\sim 7\%$  fall in quantity; to cut demand 15% you'd raise price  $\sim 11\%$ . → *Illustrates: elasticity as the design tool for a corrective ("sin") tax.*
- **World cocoa demand.** Inelastic ( $\approx -0.19$  to  $-0.96$ ; policy estimate  $-0.34$ ). → *Illustrates: inelastic demand → supply shocks hit price, so bad harvests swing incomes for growers in Ghana / Côte d'Ivoire.*
- **US beer — brand vs category (Miller et al. 2021).** Product-level elasticity  $\approx -4.74$ , category  $\approx -0.60$ . → *Illustrates: aggregation — a single brand has close substitutes (elastic), the category doesn't (inelastic).*
- **Sweden–Denmark cross-border alcohol (Asplund et al. 2007).** Swedes drive to Denmark for cheaper drink. Estimated for spirits: **own-price**  $-1.3$  (elastic), **cross-price (Danish price)**  $+0.3$  (substitute), **income**  $+1.4$  (normal good). → *Illustrates: all three elasticities at once — own, cross, income — from real data.*
- **Electricity demand (Labandeira et al. 2017).**  $\approx -0.24$  short-run vs  $-0.6$  long-run. → *Illustrates: demand is more elastic in the long run (time to adjust equipment/habits).*

## Formula sheet — quick reference

| Quantity                   | Formula                                                             | Sign / meaning                                           |
|----------------------------|---------------------------------------------------------------------|----------------------------------------------------------|
| Price elasticity of demand | $E_D = \% \Delta Q_D \div \% \Delta p$                              | <b>negative</b> ; $ E  > 1$ elastic, $ E  < 1$ inelastic |
| Point elasticity (linear)  | $E_D = (dQ/dp) \cdot (p/Q)$                                         | varies along the curve; more elastic at higher p         |
| Constant-elastic demand    | $Q = a \cdot p^\epsilon \rightarrow \ln Q = \ln a + \epsilon \ln p$ | elasticity = $\epsilon$ everywhere                       |
| Cross-price elasticity     | $E_{\text{cross}} = \% \Delta Q_A \div \% \Delta p_B$               | <b>+ substitutes, – complements</b>                      |
| Income elasticity          | $E_{\text{income}} = \% \Delta Q \div \% \Delta I$                  | <b>+ normal, – inferior</b> ; food < 1 (Engel)           |
| Supply elasticity          | $E_S = \% \Delta Q_S \div \% \Delta p$                              | <b>positive</b>                                          |
| Tax wedge                  | $p_b = p_s + t$                                                     | buyers pay more, sellers get less; DWL                   |
| Tax revenue                | $t \times Q_{\text{taxed}}$                                         | rectangle; rest of loss = deadweight loss                |
| Subsidy wedge              | $p_s = p_b + s$                                                     | expands Q above efficient; also a DWL                    |

## Exam pointers

- Compute an elasticity from two (p, Q) points and **name** it (elastic/inelastic/unit) — the –1/3 example.
- Use the point-elasticity trick  $E_D = (dQ/dp) (p/Q)$  on a **linear demand** and show elasticity **changes along the curve** (more elastic at higher price).
- Give the **sign rules** for cross-price (substitutes/complements) and income (normal/inferior) elasticities, with an example each.
- Solve a **unit-tax market** ( $p_b = p_s + t$ ) for  $p_b$ ,  $p_s$ ,  $Q$ , compute **revenue** and **deadweight loss** — the t=€3 example.
- State and **explain** the two big tax results: **legal ≠ economic incidence** (doesn't matter who's taxed) and **incidence depends on relative elasticities** (inelastic side pays more); use the sin-tax application.
- Explain **why elasticity is unit-free** and why demand is **more elastic in the long run** and for narrower product categories.

**Bridge.** Lecture 3 turns the Chapter 2 model from *qualitative* ("the curve shifts left") into *quantitative* ("by how much, and split how between price and quantity"). Elasticity is also the hinge to later chapters: **firm pricing** (a monopolist's markup depends on  $E_D$ ), **tax policy & deadweight loss** (Ch. 7), and **who gains from trade**.

## ★ Practice problems — the whole chapter (with worked solutions)

A set covering **every calculation** in this lecture. Try each closed-book first, then check against the worked solution. (Exam is open-book, no AI — so the skill being tested is doing these fast and correctly, not memorising.)

What you must be able to calculate (each maps to a problem below):

| Skill                                                              | Formula                                        | Problem |
|--------------------------------------------------------------------|------------------------------------------------|---------|
| Own-price elasticity from two points + name it                     | $E_D = \% \Delta Q \div \% \Delta p$           | 1       |
| Predict $\% \Delta Q$ from $\% \Delta p$ (and reverse)             | $\% \Delta Q = E_D \cdot \% \Delta p$          | 2       |
| Point elasticity on a linear curve (+ find the unit-elastic point) | $E_D = (dQ/dp) \cdot (p/Q)$                    | 3       |
| Elasticity → total revenue (raise or cut price?)                   | $TR = p \cdot Q$                               | 4       |
| Constant-elastic demand                                            | $Q = a \cdot p^{-\epsilon}$                    | 5       |
| Cross-price elasticity + classify                                  | $E_{cross} = \% \Delta Q_A \div \% \Delta p_B$ | 6       |
| Income elasticity + classify                                       | $E_{income} = \% \Delta Q \div \% \Delta I$    | 7       |
| Supply elasticity                                                  | $E_S = \% \Delta Q_S \div \% \Delta p$         | 8       |
| Unit tax: solve $p_b$ , $p_s$ , $Q$ , revenue, DWL, incidence      | $p_b = p_s + t$                                | 9       |
| Subsidy: solve $p_b$ , $p_s$ , $Q$ , cost, DWL                     | $p_s = p_b + s$                                | 10      |
| Incidence from elasticities (pass-through)                         | consumer share = $E_S \div (E_S +  E_D )$      | 11      |
| Equilibrium + a price control (Ch. 2 recap)                        | set $Q_D = Q_S$                                | 12      |

### Problems

1. A bakery raises bread from €4 → €5; weekly sales fall from 100 → 80 loaves. Find the price elasticity of demand and say whether demand is elastic, inelastic, or unit-elastic.
2. (a) The own-price elasticity of petrol is  $-0.3$ . If the price rises 10%, by how much does quantity demanded change? (b) A cinema wants ticket sales to rise 9%; demand elasticity is  $-1.5$ . What price change is needed?
3. Demand is  $Q = 120 - 4p$ . (a) Find  $E_D$  at  $p = 10$ . (b) Find  $E_D$  at  $p = 20$ . (c) At what price is demand unit-elastic?
4. Using the same demand  $Q = 120 - 4p$ : at  $p = 10$  and at  $p = 20$ , should the firm raise or cut price to increase total revenue? At what price is revenue maximised?

5. Demand is  $Q = 500 \cdot p^{-2}$ . (a) What is the elasticity of demand? (b) Find quantity at  $p = 5$  and  $p = 10$ . (c) If price rises 1% from €5, roughly what is  $\% \Delta Q$ ?
6. Classify each pair as substitutes, complements, or unrelated, and give  $E_{\text{cross}}$ : (a) price of **tea** +8% → demand for **coffee** +4%; (b) price of **printers** +10% → demand for **ink** +6% fall; (c) price of **umbrellas** +10% → demand for **coffee** 0%.
7. Classify the good and give  $E_{\text{income}}$ : (a) income +5% → **restaurant meals** +15%; (b) income +10% → **bus tickets** -4%; (c) income +8% → **bread** +2%.
8. When the price rises from €8 → €10, quantity supplied rises from 60 → 69. Find the price elasticity of supply and classify it.
9. A market has  $Q_D = 100 - 4p$  and  $Q_S = -20 + 2p$ . (a) Find the pre-tax equilibrium. (b) A **unit tax**  $t = €6$  is imposed. Solve for  $p_b$ ,  $p_s$ , and  $Q$ . (c) Find **tax revenue** and the **deadweight loss**. (d) How is the burden split between buyers and sellers, and **why**?
10. Same market ( $Q_D = 100 - 4p$ ,  $Q_S = -20 + 2p$ ). Now the government pays a **subsidy**  $s = €6$  per unit. Solve for  $p_b$ ,  $p_s$ ,  $Q$ , the **cost to government**, and the **deadweight loss**.
11. At the pre-tax equilibrium of problem 9, the elasticities are  $E_D = -4$  and  $E_S = +2$ . Use the pass-through formula to find the **share of the tax borne by consumers vs producers**, and check it against your answer to 9(d).
12. (Ch. 2 recap.)  $Q_D = 80 - 2p$ ,  $Q_S = -10 + p$ . (a) Find equilibrium  $p^*$  and  $q^*$ . (b) The government sets a **price ceiling** at €25. Compute the resulting **shortage**.

## Solutions

1.  $\% \Delta Q = (80 - 100)/100 = -20\%$ ;  $\% \Delta p = (5 - 4)/4 = +25\%$ .

$$E_D = -20\% \div +25\% = -0.8$$

$|-0.8| < 1 \rightarrow$  **inelastic** (quantity moves less than price). \* (Note: revenue actually rises — see problem 4's logic.)\*

2. (a)  $\% \Delta Q = E_D \cdot \% \Delta p = -0.3 \times 10\% = -3\%$  (quantity falls 3%). (b) rearrange:  $\% \Delta p = \% \Delta Q \div E_D = 9\% \div (-1.5) = -6\% \rightarrow$  the cinema must **cut** price 6%.

3. Slope  $dQ/dp = -4$  (constant). Use  $E_D = (dQ/dp) \cdot (p/Q)$ .

- (a)  $p = 10 \rightarrow Q = 120 - 40 = 80 \rightarrow E_D = -4 \times 10/80 = -0.5$  (inelastic).
- (b)  $p = 20 \rightarrow Q = 120 - 80 = 40 \rightarrow E_D = -4 \times 20/40 = -2$  (elastic).
- (c) unit-elastic: set  $-4 \cdot (p/Q) = -1 \rightarrow Q = 4p \rightarrow 120 - 4p = 4p \rightarrow 8p = 120 \rightarrow p = 15$  ( $Q = 60$ ). (On any straight-line demand, the unit-elastic point is the **midpoint**.)

4. Revenue  $TR = p \cdot Q$ .

- At  $p = 10$  (inelastic,  $|E| < 1$ ): raising price **raises** revenue. Check:  $p=10 \rightarrow TR = 10 \times 80 = 800$ ;  $p=11 \rightarrow Q = 76$ ,  $TR = 836$ .  $\rightarrow$  **raise price**.

- At  $p = 20$  (elastic,  $|E| > 1$ ): cutting price **raises** revenue. Check:  $p=20 \rightarrow TR = 800$ ;  $p=19 \rightarrow Q = 44$ ,  $TR = 836$ .  $\rightarrow$  **cut price**.
- **Revenue is maximised at the unit-elastic point**  $p = 15$ :  $TR = 15 \times 60 = 900$ . (Rule:  $|E| < 1 \rightarrow$  raise price;  $|E| > 1 \rightarrow$  cut price; max TR where  $|E| = 1$ .)

5. (a) For  $Q = a \cdot p^\epsilon$ , the elasticity is the exponent:  $E_D = -2$  (constant at every price). (b)  $p = 5 \rightarrow Q = 500/25 = 20$ ;  $p = 10 \rightarrow Q = 500/100 = 5$ . (c)  $\% \Delta Q \approx E_D \cdot \% \Delta p = -2 \times 1\% = -2\%$ .

6.  $E_{cross} = \% \Delta Q_A \div \% \Delta p_B$ . (a)  $+4 \div +8 = +0.5 \rightarrow$  **substitutes** (tea dearer  $\rightarrow$  more coffee). (b)  $-6 \div +10 = -0.6 \rightarrow$  **complements** (printers dearer  $\rightarrow$  less ink). (c)  $0 \div +10 = 0 \rightarrow$  **unrelated**.

7.  $E_{income} = \% \Delta Q \div \% \Delta I$ . (a)  $+15 \div +5 = +3 \rightarrow$  **normal, and a luxury** ( $E > 1$ ). (b)  $-4 \div +10 = -0.4 \rightarrow$  **inferior**. (c)  $+2 \div +8 = +0.25 \rightarrow$  **normal necessity** ( $0 < E < 1$ ; cf. Engel's law for food).

8.  $\% \Delta Q_S = (69 - 60)/60 = +15\%$ ;  $\% \Delta p = (10 - 8)/8 = +25\%$ .

$$E_S = +15\% \div +25\% = +0.6$$

$0 < E_S < 1 \rightarrow$  **inelastic supply** (output hard to expand quickly).

9. (a) **Pre-tax**: set  $Q_D = Q_S$ :  $100 - 4p = -20 + 2p \rightarrow 120 = 6p \rightarrow p^* = 20, Q^* = 20$ .

- (b) **With  $t = 6$** , use  $p_b = p_s + 6$ ,  $Q_D = 100 - 4p_b$ ,  $Q_S = -20 + 2p_s$ . Set equal:  $100 - 4(p_s + 6) = -20 + 2p_s \rightarrow 76 - 4p_s = -20 + 2p_s \rightarrow 96 = 6p_s \rightarrow p_s = 16, p_b = 22, Q = 100 - 4(22) = 12$ . (check  $Q_S = -20 + 32 = 12 \checkmark$ )
- (c) **Tax revenue** =  $t \times Q = 6 \times 12 = \text{€}72$ . **DWL** =  $\frac{1}{2} \times t \times (Q^* - Q_{tax}) = \frac{1}{2} \times 6 \times (20 - 12) = \text{€}24$ .
- (d) Buyers bear  $p_b - p^* = 22 - 20 = \text{€}2$ ; sellers bear  $p^* - p_s = 20 - 16 = \text{€}4$  (they sum to  $t = 6$ ). **Sellers bear more because supply is the more inelastic side** (slope 2 vs demand's 4  $\rightarrow$  demand is more elastic, so it escapes more of the tax onto quantity).

10. **With  $s = 6$** , use  $p_s = p_b + 6$ :  $100 - 4p_b = -20 + 2(p_b + 6) \rightarrow 100 - 4p_b = -8 + 2p_b \rightarrow 108 = 6p_b \rightarrow p_b = 18, p_s = 24, Q = 100 - 4(18) = 28$ . (check  $Q_S = -20 + 48 = 28 \checkmark$ ) — buyers pay less ( $18 < 20$ ), sellers get more ( $24 > 20$ ), and **Q rises** ( $20 \rightarrow 28$ ). **Cost to government** =  $s \times Q = 6 \times 28 = \text{€}168$ . **DWL** =  $\frac{1}{2} \times s \times (Q_{sub} - Q^*) = \frac{1}{2} \times 6 \times (28 - 20) = \text{€}24$  (over-production: units that cost more than they're worth).

11. Pass-through shares:

#### INCIDENCE FROM ELASTICITIES

$$\text{consumer share} = E_S \div (E_S + |E_D|) \quad \cdot \quad \text{producer share} = |E_D| \div (E_S + |E_D|)$$

With  $E_S = 2$  and  $|E_D| = 4$ : **consumer share** =  $2 \div (2+4) = 1/3$ , **producer share** =  $4 \div 6 = 2/3$ . Of the €6 tax that's **€2 on buyers, €4 on sellers** — exactly matching 9(d).  $\checkmark$  (Rule: the **more inelastic** side pays the bigger share.)

12. (a)  $80 - 2p = -10 + p \rightarrow 90 = 3p \rightarrow p^* = 30, q^* = 20$ . (b) At the ceiling  $p = 25$ :  $Q_D = 80 - 50 = 30$ ,  $Q_S = -10 + 25 = 15 \rightarrow$  **shortage** =  $Q_D - Q_S = 30 - 15 = 15$  units.

## ★ Fun facts & memorable details (Lecture 3)

Sticky bits from Elasticities & Taxes.

- **It doesn't matter who you tax.** In a competitive market the burden of a tax is **identical** whether the government bills the shop or the shopper — "all that matters is that the tax drives a wedge." (This is why arguing about *who* a tax is "on" often misses the point.)
- **Why cigarette & alcohol taxes raise so much money:** demand is **inelastic**, so people keep buying — the tax is **passed to consumers** with little drop in quantity (and is often **regressive**).
- **The soda-tax number:** the own-price elasticity of sugary drinks ( $\approx -1.37$ ) is a real input to real policy — a 5% price rise cuts consumption  $\sim 7\%$ .
- **Engel's law (1857):** the **richer** you get, the **smaller** the share of income you spend on **food** — one of the oldest empirical regularities in economics, and still used to gauge living standards.
- **Pharma = "defensive stock":** demand for medicine barely moves with income (income-inelastic), so drug companies hold up in recessions.
- **Elasticity has a time dimension:** when petrol jumps, you can't do much *this week* (inelastic), but over years you buy a smaller car and move closer to work (elastic) — the "short-run vs long-run" split.
- **Brand vs category:** demand for *a* beer is wildly elastic ( $\approx -4.7$  — switch brands!), but demand for *beer* is inelastic ( $\approx -0.6$ ) — same drink, opposite elasticity, just different zoom level.
- **Swedes really do drive to Denmark for cheap booze** — enough that economists measured the cross-border **cross-price elasticity** of spirits ( $+0.3$ ) from the data (Asplund et al. 2007).

## Lecture 4: Consumer Choice (Chapter 5)

**Required reading:** Friberg, **Chapter 5 — Consumer choice** (§5.1 modelling choice · §5.2 preferences & utility · §5.3 budget constraints · §5.4 utility maximization · §5.5 discussion of the framework · §5.6 behavioral economics · §5.7 appendix: solving for the optimal basket with Lagrange). *Lecturer: Luigi Butera — deck "Lecture 4: Consumer choice."* (Numbering note: this is the course's **4th lecture** but the textbook's **Chapter 5**; the deck's own "4.1–4.5" roadmap just re-numbers the chapter's sections.)

**Theme of the lecture (Butera — from the deck).** So far (Lectures 2–3) we took **demand curves as given** and pushed them around. Now we **build demand from the ground up**: *where do preferences come from, and how does a rational consumer choose under scarcity?* The deck's answer is a **three-stage machine** — **preferences → utility → budget constraint → constrained optimization** — that will (next chapter) *generate* the demand curve. The deck's five stops: **4.1 preferences · 4.2 utility · 4.3 budget constraints · 4.4 constrained choice · 4.5 behavioral economics**.

★ **The heart of the lecture:** a consumer **maximizes utility subject to a budget constraint**. We (1) describe what she **likes** with well-behaved **preferences** (complete + transitive), (2) quantify them with a **utility function** whose slope-of-indifference-curve is the **MRS** = how much of one good she'll trade for the other *and stay equally happy*, (3) describe what she **can afford** with a **budget line** whose slope is the **MRT** =  $p_F/p_E$  = how the market lets her trade the goods. The **optimum** is

where the highest reachable indifference curve is **tangent** to the budget line — i.e. where  $MRS = MRT$  ( $MU_F/MU_E = p_F/p_E$ ): the rate she's *willing* to trade equals the rate the market *lets* her trade.

**The red thread.** Every piece answers one of three questions: *what do I want?* (preferences → utility → MRS), *what can I afford?* (budget line → MRT), and *so what do I pick?* (tangency:  $MRS = MRT$ ). Get those three and you can predict how choices shift when **prices** or **income** change — which is exactly a **demand curve** (next chapter). §5.6 then asks: *when does this rational machine break down?* (behavioral economics).

## Key concepts / "modes" to use

- **Bundle** — a list of quantities of each good (e.g. "4 Entertainment, 2 Food"). Choice = ranking bundles.
- **Preferences & the rationality axioms** — **completeness** (can rank any two bundles) • **transitivity** (rankings "hold in transit") • **monotonicity** / "**more is better**" (convenient add-on).
- **Utility function**  $U(F, E)$  — assigns a number to each bundle so that **higher = preferred**. **Ordinal**, not cardinal (only the *ranking* means anything).
- **Marginal utility (MU)** — extra utility from one more unit of a good, holding the other fixed = the **partial derivative**; typically **diminishing**.
- **Indifference curve** — all bundles giving the *same* utility; **5 properties**; three shapes (**perfect substitutes**, **imperfect substitutes**, **perfect complements**).
- **Marginal rate of substitution (MRS)** — slope of the indifference curve =  $MU_F / MU_E$  = how much E you'll give up for one more F at equal utility.
- **Budget constraint / line** —  $I = p_F \cdot F + p_E \cdot E$ ; the **opportunity set** is everything on or under it.
- **Marginal rate of transformation (MRT)** — slope of the budget line =  $p_F / p_E$  = how the *market* lets you swap the goods.
- **Consumer optimum** — highest indifference curve tangent to the budget line:  $MRS = MRT$ , i.e.  $MU_F/MU_E = p_F/p_E$  (equivalently, equal marginal utility *per euro*:  $MU_F/p_F = MU_E/p_E$ ).
- **Interior vs corner solution** — tangency inside the axes vs "spend everything on one good" (perfect substitutes).
- **Behavioral economics** — systematic deviations: **social preferences**, **lack of self-control**, **cognitive biases** (loss aversion, endowment effect, framing, overconfidence).

## Section-by-section main points

### §5.1 How can we model choice? — the three ingredients (deck)

**Core:** we model an individual who **faces constraints** (income, time) and **does as well as possible** given them. Sounds almost tautological — the whole chapter is about *putting structure* on "does as well as possible." The deck lists the **ingredients we need**:

- A language to say "I like this more than that" → **preferences** (§5.2).
- **Assumptions** that make those preferences *rational* → the **axioms** (completeness, transitivity).
- A tool to quantify wellbeing → the utility function (§5.2.1).
- A tool to quantify "you can't have it all" → the **budget constraint** (§5.3).

- Bringing it together → constrained optimization (§5.4).
- Three premises of the whole build (deck + §5.1): (1) we know and have stable preferences, (2) we face constraints (time, money), (3) we try to do as well as possible given them.

## §5.2 Preferences and the rationality axioms (deck slides 6–7)

**Core:** choice is modelled as **ranking bundles**. For the ranking to give a consistent model we assume it obeys two **essential** axioms (plus one convenient one). The deck's notation for the ranking:

| Symbol        | Reads                                 | Meaning                           |
|---------------|---------------------------------------|-----------------------------------|
| $a \succ b$   | "a is <b>strictly preferred</b> to b" | a is <i>better</i> than b         |
| $a \succeq b$ | "a is <b>weakly preferred</b> to b"   | a is <i>at least as good</i> as b |
| $a \sim b$    | " <b>indifferent</b> between a and b" | a and b are <i>equally good</i>   |

- **Completeness (essential):** the consumer can **rank any two bundles** — for any A, B she can say  $A \succ B$ ,  $B \succ A$ , or  $A \sim B$ . (No "I can't compare these.")
- **Transitivity (essential):** rankings "**hold in transit**" — if  $A \succ B$  and  $B \succ C$ , then  $A \succ C$ . This is what lets us predict choice in a *chain* of comparisons (and stops preferences from cycling).
- **Monotonicity / "more is better" (convenient, not essential):** all else equal, **more of a good is better**. This is what makes bundles *farther from the origin* preferred. (*Things you want less of — pollution, noise — are relabelled "bads," and we redefine the good as their absence.*)
- **Why the axioms matter:** completeness + transitivity are *exactly* the conditions under which preferences can be represented by a utility function (next section). No axioms → no utility function → no model.

### §5.2.1 Utility & utility functions (deck slides 8–9)

**Core:** if preferences are **complete and transitive**, we can represent them with a utility function  $U(\cdot)$  that assigns each bundle a **number**, higher for more-preferred bundles. Neoclassical economics inherits this from the **utilitarian** tradition (J.S. Mill: "*actions are right in proportion as they tend to promote happiness*").

UTILITY REPRESENTS PREFERENCES

$$a \succ b \iff U(a) > U(b)$$

**Reads:** the utility function just **translates the ranking into real numbers** — bundle *a* is preferred to *b* if and only if *a* scores a higher utility number. (This is the **von Neumann–Morgenstern** representation the deck cites.) The function *orders* preferences; it does not measure "happiness" in any absolute unit.

- **Ordinal, not cardinal (the crucial caveat):** only the **ranking** matters, **not** the size of the gap.  $U = 8$  vs  $U = 4$  means the first is **preferred** — it does **not** mean "twice as good." (Deck: " $9 > 1$ , but do '1' and '9' mean something meaningful?" — no. *Utility is unobservable.*)
- **General vs specific form:** general  $U = U(F, E)$ ; a specific example the textbook uses is  $U = \sqrt{EF}$ . **Worked:** at  $E=2$ ,  $F=8$ ,  $U = \sqrt{(2 \cdot 8)} = \sqrt{16} = 4$ . Bundles with the **same product**  $E \cdot F = 16$  (e.g.

E=4,F=4 or E=8,F=2) sit on the same indifference curve; E=5,F=5 gives  $U = 5 \rightarrow$  a **higher** curve.

- **Common functional forms (deck slide 12):**

- **Cobb-Douglas:**  $U = F^a \cdot E^b$  — the workhorse; convex, imperfect substitutes.
- **Perfect substitutes:**  $U = a \cdot F + b \cdot E$  — straight-line indifference curves.
- *(Perfect complements:  $U = \min(F, E)$  — L-shaped.)*

## ★ Marginal utility & diminishing marginal utility (deck slides 10, 16)

**Core:** marginal utility (MU) is the extra utility from **one more (infinitesimal) unit** of a good, holding the others fixed — i.e. the **partial derivative** of  $U$ . It is usually **diminishing** (the 1st slice of pizza beats the 7th).

### MARGINAL UTILITY

$$MU_F = \partial U(F,E)/\partial F \quad \cdot \quad MU_E = \partial U(F,E)/\partial E$$

*Reads:* **how much total utility rises per extra unit of one good, other goods held constant** —  $\partial U/\partial F$  = the partial derivative w.r.t. Food • **diminishing MU** = this derivative *shrinks* as you get more of the good.

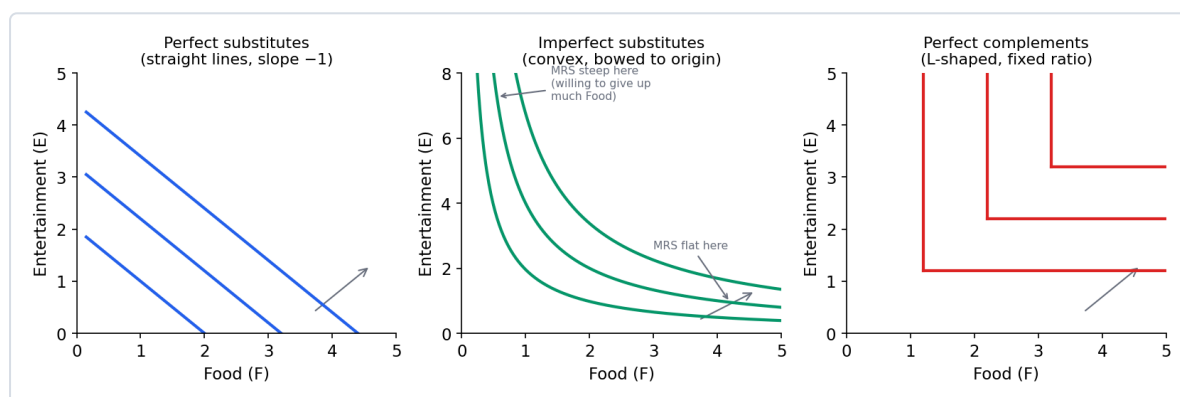
- **The deck's three consumers (utility over one good):** for each extra apple —
  - **Bob:** each apple adds the **same** utility  $\rightarrow$  **constant** MU (linear U).
  - **Linda:** each apple adds a bit **more**  $\rightarrow$  **increasing** MU (convex U). (*unusual*)
  - **Mike:** each apple adds a bit **less**  $\rightarrow$  **diminishing** MU (concave U). (*the normal case*)
- **Worked example ( $U = \sqrt{EF} = E^{0.5}F^{0.5}$ ):**  $MU_F = \partial U/\partial F = 0.5 \cdot E^{0.5} \cdot F^{-0.5} = 0.5 \cdot \sqrt{E/F}$ . At E=2, F=8:  $MU_F = 0.5 \cdot \sqrt{2/8} = 0.5 \cdot 0.5 = 0.25$ . As F rises (holding E),  $\sqrt{E/F}$  falls  $\rightarrow$   **$MU_F$  falls**  $\rightarrow$  diminishing marginal utility. ✓
- **Why it matters:** diminishing MU is *why* indifference curves are **bowed toward the origin** (convex) and *why* the **MRS falls** as you move down a curve — the next two ideas.

## Indifference curves — 5 properties & the 3 types (deck slides 13–14)

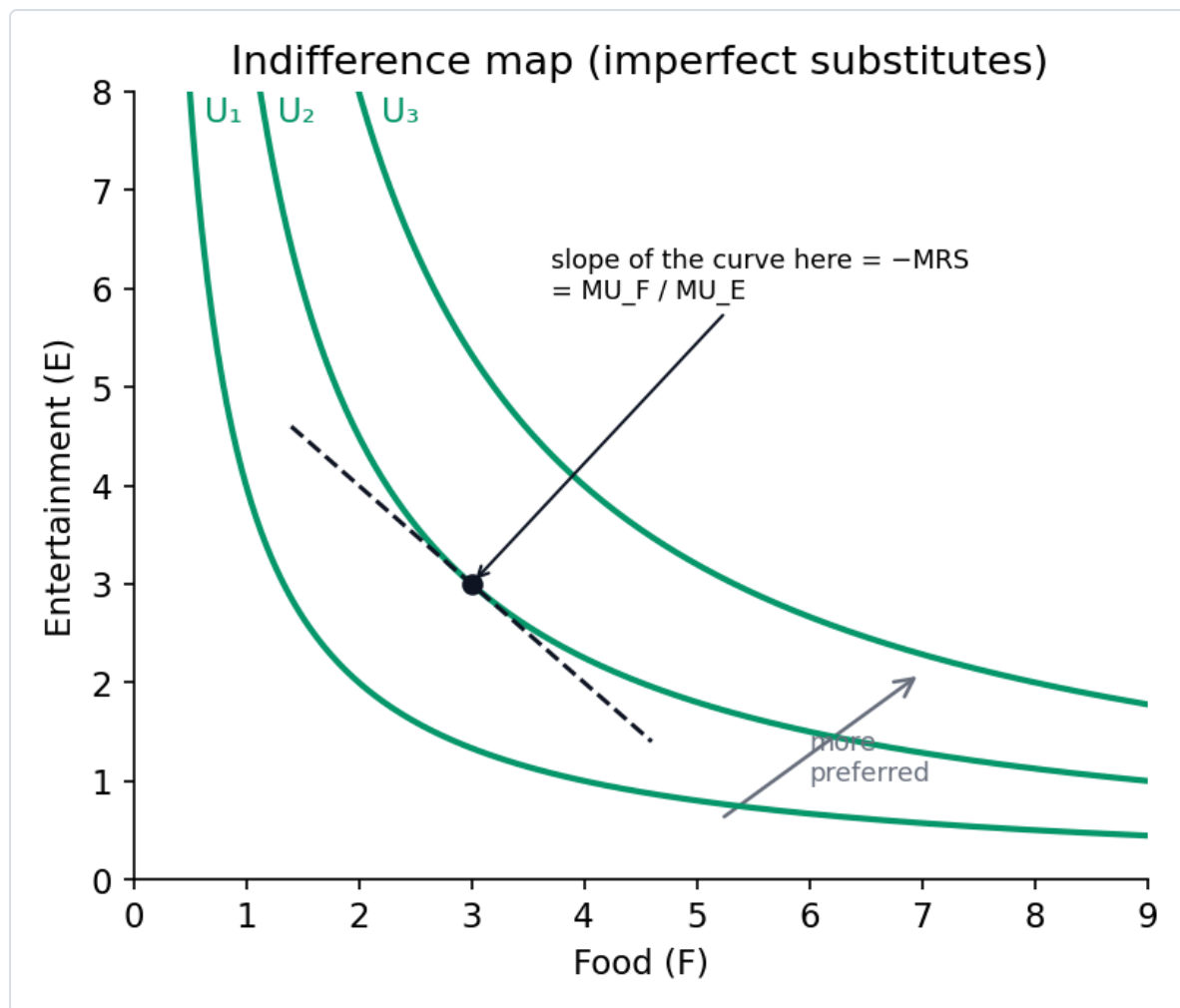
**Core:** an indifference curve connects **all bundles the consumer views as equally desirable**. It's the level-set of the utility function (fix  $U$  at one number). The deck's **five properties**:

| # | Property                                                   | Why                                                                |
|---|------------------------------------------------------------|--------------------------------------------------------------------|
| 1 | Curves <b>farther from the origin</b> are preferred        | "more is better" (monotonicity)                                    |
| 2 | <b>Every</b> bundle lies on some <u>indifference curve</u> | completeness                                                       |
| 3 | Indifference curves <b>cannot cross</b>                    | transitivity (crossing → contradiction)                            |
| 4 | They <b>slope downward</b>                                 | to keep utility fixed, more of one good ⇒ less of the other        |
| 5 | They <b>cannot be "thick"</b>                              | a thick band would contain a strictly-better bundle (monotonicity) |

- (Utility is read **ordinally**: the labels on the curves rank them; the gaps between labels are meaningless.)
- Three shapes of preference (textbook §5.2):



- **Perfect substitutes** — tradeable at a **constant rate** → **straight-line** curves (constant slope). *Example: one €20 note ~four €5 notes, always.*  $U = aF + bE$ .
- **Imperfect substitutes (the normal case)** — willing to trade, but the **rate depends on how much of each you already have** → **convex** curves, "bowed toward the origin" (**strictly convex** preferences). *When you have lots of chips and no drink, you'll give up many chips for one drink; when you have few chips, you won't.*  $U = \sqrt{EF}$ , Cobb-Douglas.
- **Perfect complements** — consumed in a **fixed ratio** → **L-shaped** curves. *Left & right shoes: a 2nd right shoe alone adds nothing.*  $U = \min(F, E)$ .



**How to read it (indifference map).** Each green curve is one utility level; **northeast = more preferred** (property 1). The dashed tangent at the marked bundle has slope  $-MRS$ : it gets **flatter** as you move right (more Food, less willing to give up Food) — that's diminishing MRS, the topic below.

## ★ Marginal rate of substitution (MRS) (deck slides 17–19)

**Core:** the **MRS** is the slope of the indifference curve — the maximum amount of one good the consumer will give up to get one more unit of the other, staying equally happy. Because utility is constant along the curve ( $dU = 0$ ), it equals the **ratio of marginal utilities**.

### MARGINAL RATE OF SUBSTITUTION

$$MRS = -dE/dF = MU_F / MU_E \quad (\text{good on the x-axis over good on the y-axis})$$

**Reads:** how much E you'll trade for one more F and stay on the same curve — derive it by **totally differentiating**  $U$  and setting  $dU = 0$ :  $0 = MU_F \cdot dF + MU_E \cdot dE \rightarrow -dE/dF = MU_F / MU_E$ .  $MU_F$  = marginal utility of the x-axis good (Food) •  $MU_E$  = marginal utility of the y-axis good (Entertainment) • the ratio = the curve's steepness at that point.

- **Diminishing MRS (why the curve bows):** as you move **down-right** along the curve (more F, less E),  $MU_F$  falls and  $MU_E$  rises, so the ratio  $MU_F / MU_E$  **shrinks** — the curve **flattens**. "When I have relatively less pizza I'll give up more burritos for a slice than when I already have lots of pizza."

- **Worked example ( $U = \sqrt{EF}$ ):**  $MU_F = 0.5\sqrt{E/F}$ ,  $MU_E = 0.5\sqrt{F/E}$ , so  $MRS = MU_F/MU_E = (0.5\sqrt{E/F})/(0.5\sqrt{F/E}) = E/F$ . At bundle  $E=2, F=8$ :  $MRS = 2/8 = 0.25 \rightarrow$  she'll give up only **0.25 units of E** for one more F (she already has lots of F). At  $E=8, F=2$ :  $MRS = 8/2 = 4 \rightarrow$  now she'll give up **4 units of E** for one more F. *Same curve, opposite ends  $\rightarrow$  diminishing MRS.* ✓

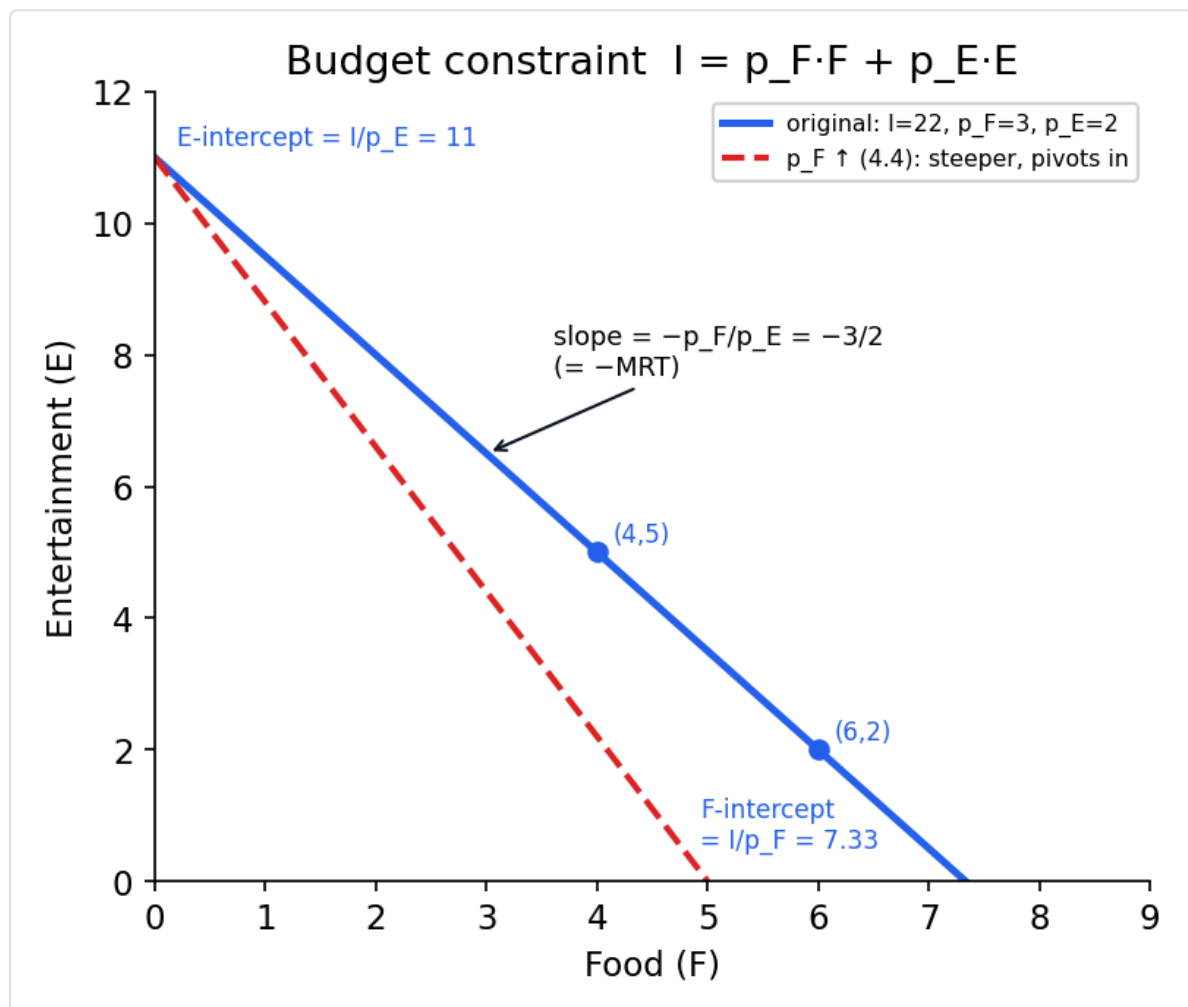
## §5.3 Budget constraint & the MRT (deck slides 21–22)

**Core:** the **budget constraint** operationalizes "you can't have it all": given prices and income  $I$ , total spending can't exceed income. With two goods it's a **straight line** (the **budget line**); everything on or below it is the **opportunity set**.

### BUDGET CONSTRAINT & LINE

$$I = p_F F + p_E E \implies E = I/p_E - (p_F/p_E) \cdot F$$

*Reads:* income = spending on Food + spending on Entertainment; rearranged for graphing, E-intercept =  $I/p_E$  (all income on E), slope =  $-p_F/p_E$ , F-intercept =  $I/p_F$  (all income on F).



- **Worked example (deck:  $I=22, p_F=3, p_E=2$ ):** E-intercept =  $22/2 = 11$ ; slope =  $-3/2$ ; F-intercept =  $22/3 \approx 7.33$ . Bundle  $(F=4, E=5)$  is affordable:  $3 \cdot 4 + 2 \cdot 5 = 22$  ✓. So is  $(F=6, E=2)$ :  $3 \cdot 6 + 2 \cdot 2 = 22$  ✓.

- **MRT — marginal rate of transformation** = the **slope of the budget line** = how the **market** lets you swap goods:

#### MARGINAL RATE OF TRANSFORMATION

$$MRT = p_F / p_E \quad (\text{the relative price = the market "exchange rate" between the goods})$$

*Reads:* give up one F, the market gives you  $p_F/p_E$  units of E. With  $p_F=3$ ,  $p_E=2$ ,  $MRT = 1.5$ : one less Food frees €3, which buys 1.5 Entertainment. (*MRS = what you're willing to trade; MRT = what the market lets you trade.*)

- **Comparative statics of the line (deck):**
  - **Income change** → **parallel shift** (slope unchanged; both intercepts move).
  - **Price change** → **pivot** (steepness changes). A higher  $p_F \Rightarrow$  **steeper**, pivoting **inward** around the E-intercept (see red dashed line); a lower  $p_F \Rightarrow$  **flatter**.

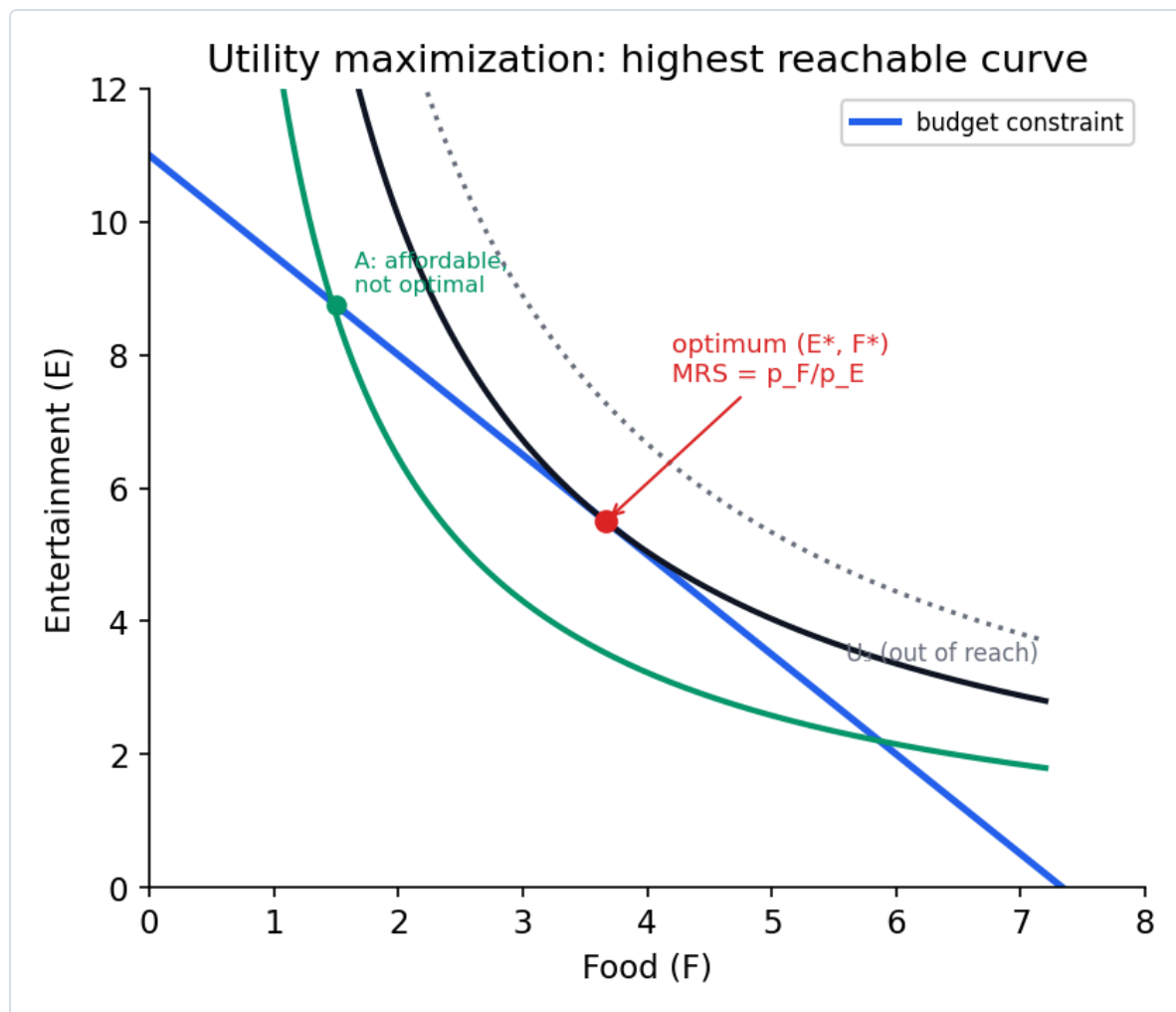
### ★ §5.4 The optimum: maximize utility subject to the budget (deck slide 23)

*Core:* the consumer solves **max  $U(F,E)$  subject to  $p_F \cdot F + p_E \cdot E \leq I$** . Graphically: find the **highest indifference curve** that still touches the budget line — that's the **tangency point**. At the tangency, the slope of the indifference curve (**MRS**) equals the slope of the budget line (**MRT**).

#### CONSUMER OPTIMUM (TANGENCY)

$$MRS = MRT \iff MU_F/MU_E = p_F/p_E \iff MU_F/p_F = MU_E/p_E$$

*Reads:* at the optimum, **the rate you're willing to trade = the rate the market lets you trade**. The last form is the "**equal marginal utility per euro**" (bang-for-the-buck) rule: **the last euro spent on Food yields the same extra utility as the last euro spent on Entertainment** — otherwise you'd reallocate toward the higher-MU-per-euro good.



**How to read it (utility maximization).** Point A is *affordable* (on the budget line) but **not optimal**: the curve through A is steeper than the line ( $MRS > MRT$ ), so she values Food more than the market charges → she should **buy more Food, less Entertainment**, sliding down to the tangency. The **red optimum** is the best reachable bundle; the dotted curve  $U_3$  would be better but her income can't reach it.

- **Why A isn't optimal (the deck's logic):** at A,  $MU_F/MU_E > p_F/p_E$  — food is "cheap relative to how much you value it." Reallocate toward food until the two ratios are equal.
- **Worked example ( $U = \sqrt{EF}$ ),  $I=22$ ,  $p_F=3$ ,  $p_E=2$ ):**  $MRS = E/F$ . Set  $MRS = MRT$ :  $E/F = 3/2$  →  $E = 1.5F$ . Plug into the budget  $3F + 2E = 22$ :  $3F + 2(1.5F) = 22$  →  $6F = 22$  →  $F^* = 3.67$ ,  $E^* = 5.5$ . (Check:  $3 \cdot 3.67 + 2 \cdot 5.5 = 11 + 11 = 22$  ✓ — matches the red dot in the figure.)

## Comparative statics · interior vs corner · the Cobb-Douglas shortcut (deck slide 24, §5.7)

**Core:** once you can find the optimum, you can ask **how it moves** when a price or income changes — and that movement, traced out, **is the demand curve** (Ch. 6). Watch for **corner solutions**.

- **Interior vs corner solution (deck):**
  - **Interior solution** — the tangency lies **strictly inside** the axes (buy *some* of both goods). Requires **convex** (imperfect-substitute) preferences.
  - **Corner solution** — the optimum is **on an axis** (buy *only one* good). Happens with **perfect substitutes**: if  $MRS \neq MRT$  everywhere, you spend **everything** on whichever good gives more

utility per euro ( $MU/p$ ). Example: if  $U = 2F + E$  and  $p_F = p_E$ , Food gives twice the bang per euro  $\rightarrow$  buy only Food.

- **Perfect complements** — always a corner-of-the-L: buy the goods in their **fixed ratio** regardless of the price ratio (solve the ratio + the budget line together).
- **The Cobb-Douglas shortcut (worth memorising for exams):** for  $U = F^a \cdot E^b$ , the tangency condition solves in closed form — the consumer **spends a constant share of income on each good**:

#### COBB-DOUGLAS OPTIMAL DEMANDS

$$F^* = \left[ \frac{a}{a+b} \right] \cdot I/p_F \quad \cdot \quad E^* = \left[ \frac{b}{a+b} \right] \cdot I/p_E$$

*Reads:* spend the fraction  $a/(a+b)$  of income on Food and  $b/(a+b)$  on Entertainment, then divide by each price to get quantities. The exponents are the expenditure shares. **Worked ( $U = F^{0.5}E^{0.5}$ )**, so  $a=b=0.5$ ,  $I=22$ ,  $p_F=3$ ,  $p_E=2$ :  $F^* = 0.5 \cdot 22/3 = 3.67$ ,  $E^* = 0.5 \cdot 22/2 = 5.5$  — same answer as the tangency method above, in one line. (Notice each good's demand **doesn't depend on the other good's price** — a special Cobb-Douglas feature.)

- **The appendix method (§5.7) — Lagrange:** for a general  $U$ , combine objective + constraint into  $\mathcal{L} = U(F,E) + \lambda(I - p_F F - p_E E)$ , set the three partials to zero, and the first two give back  $MU_F/MU_E = p_F/p_E$  — the same tangency condition.  $\lambda$  = the **marginal utility of income** ("how much extra utility one more euro of income buys"). *You won't need Lagrange for the exam, but know that it reproduces  $MRS = MRT$ .*
- **The payoff (Ch. 6 foreshadow):** change  $p_F$ , re-solve the optimum, and record the new  $F^*$  — repeat, and the  $(p_F, F^*)$  pairs **trace out the demand curve**. *This is why we built all this machinery: choice  $\rightarrow$  demand.*

## §5.5–5.6 Discussion of the framework & behavioral economics

**Core:** the rational-choice model is powerful (Becker: "maximizing behavior, market equilibrium, and stable preferences, used relentlessly and unflinchingly") — but it's a **model**. Two clarifications, then the **deviations**.

- **Two common misconceptions (§5.5):**
  - "Economics says be a selfish materialist." **No** — the framework maximizes **utility**, not **goods**. Giving to charity, choosing a low-paid academic job you love, buying organic — all perfectly rational if that's your preference.
  - "It only works for apples-vs-oranges." Actually **Gary Becker** extended it to fertility, marriage, crime, discrimination: assume **stable preferences** and let **costs/benefits differ** across contexts (e.g. why fertility differs across countries — Table 5.2 — need not be different *tastes*).
- **Behavioral economics (§5.6) — three deviations from the simple model:**

| Deviation                                 | What it is                                                                                                        | Signature examples                                                                                                                                                                                                                  |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Social preferences</b>                 | We care about <b>others</b> — fairness, status, altruism, spite. <i>Not irrational</i> — just richer preferences. | <b>Ultimatum game</b> : proposers offer ~40% (not ~0%), responders reject ~16% of low offers to punish unfairness.                                                                                                                  |
| <b>Lack of self-control</b>               | We over-weight the <b>present</b> — a myopic " <i>doer</i> " vs a " <i>planner</i> " (Thaler–Shefrin).            | Smoking, over-spending, under-saving; the marshmallow test; policy responses = mandatory pensions, alcohol limits.                                                                                                                  |
| <b>Cognitive limitations &amp; biases</b> | Systematic (not random) errors; <b>bounded rationality</b> , heuristics.                                          | <b>Loss aversion / status-quo bias / endowment effect</b> (Cornell coffee-mug study: owners value the mug 2–4× more); <b>framing</b> (600-lives problem); <b>overconfidence</b> (68% of entrepreneurs think they'll beat the odds). |

- **Random error vs bias (deck's dartboard)**: *random* misses scatter around the bull's-eye (people are right *on average*); a **behavioral bias** is a **systematic** skew off-centre (Kahneman & Tversky, *Thinking, Fast and Slow*).
- **Final word (§5.6.5)**: the rational model stays the **benchmark** — you need it *even to define* a "deviation." Behavioral economics tells you **when** the benchmark will mislead (e.g. demand for addictive painkillers vs baking soda).

## Cases & examples

- **Why did shrimp consumption triple? (Freakonomics / Levitt) — §5.4.1**. Non-economists guessed **demand shifts** (health fads, movies, trendy chefs). The choice framework points first to **prices and income**: the **relative price of shrimp fell** — that alone moves the optimum along/across the budget line. *Lesson: shifts in the budget constraint, not just tastes, drive choices.*
- **Same preferences, different choices: US pizza vs French duck confit (Dubois et al. 2014) — Fig 5.8**. Give an American and a French consumer the **identical** indifference map. Pizza is **cheaper in the US**, confit **cheaper in France** → different **budget-line slopes** → the American picks more pizza, the French more confit. *Different choices need not mean different tastes — just different relative prices.*
- **The ultimatum game — fairness is real (Güth 1982; Henrich 2001)**. Split €10; responder can veto (then both get €0). Pure self-interest predicts "offer €0.10, accept." Instead proposers offer **~40%**, and low offers get **rejected** — even across 15 small-scale societies worldwide (offers never below ~25%). *Evidence for social preferences.*
- **The Cornell coffee-mug experiment — the endowment effect (Kahneman et al. 1990)**. Mugs handed out at random; owners then valued them **2–4× higher** than non-owners would pay. Ownership shouldn't matter to a rational valuation — but a **reference point** does. *Loss aversion / status-quo bias in one image.*
- **Food stamps or cash? (deck slide 25)**. ~50% of US kids use food stamps before age 20. In the standard model, **cash is weakly better** (it relaxes the budget line *more* — you can buy food *or* anything else), so an in-kind transfer can only tie or lose. Yet food stamps persist — because of **paternalism / self-control** concerns and political constraints. *A perfect budget-constraint application with a behavioral twist.*

- **Fertility rates: tastes or costs? (Becker; Table 5.2).** South Korea 0.97 kids/woman vs Mali 6.14. A Becker-style reading: **same preferences**, different **costs/benefits** of children (childcare compatibility with careers, old-age support, child mortality, pensions). *Rational choice reaches far beyond "goods."*

## Formula sheet (Consumer Choice)

| Concept                                          | Formula                                                                             | In words                                                 |
|--------------------------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------|
| <b>Utility represents preferences</b>            | $a > b \Leftrightarrow U(a) > U(b)$                                                 | numbers rank bundles ( <b>ordinal</b> )                  |
| <b>Marginal utility</b>                          | $MU_F = \partial U / \partial F$                                                    | extra utility per extra unit; usually <b>diminishing</b> |
| <b>MRS (slope of indifference curve)</b>         | $MRS = MU_F / MU_E$                                                                 | how much E you'll trade for one more F, same utility     |
| <b>Budget line</b>                               | $I = p_F \cdot F + p_E \cdot E \rightarrow E = I/p_E - (p_F/p_E)F$                  | what you can afford; slope $-p_F/p_E$                    |
| <b>MRT (slope of budget line)</b>                | $MRT = p_F / p_E$                                                                   | how the <b>market</b> lets you trade F for E             |
| <b><u>Consumer optimum</u></b>                   | $MRS = MRT \Leftrightarrow MU_F/MU_E = p_F/p_E \Leftrightarrow MU_F/p_F = MU_E/p_E$ | tangency = equal MU per euro                             |
| <b>Cobb-Douglas demands</b><br>( $U = F^a E^b$ ) | $F^* = a/(a+b) \cdot I/p_F$ , $E^* = b/(a+b) \cdot I/p_E$                           | spend constant income shares                             |

## Key terms

| Term                                    | One-line meaning                                                         |
|-----------------------------------------|--------------------------------------------------------------------------|
| <b>Bundle</b>                           | a specific quantity of each good                                         |
| <b>Completeness</b>                     | can rank <i>any</i> two bundles                                          |
| <b>Transitivity</b>                     | $A > B, B > C \Rightarrow A > C$ (rankings "hold in transit")            |
| <b>Monotonicity</b>                     | "more is better" — farther-out curves preferred                          |
| <b>Utility function</b>                 | numbers on bundles preserving the ranking ( <b>ordinal</b> )             |
| <b>Marginal utility (MU)</b>            | $\partial U / \partial (\text{good})$ ; extra utility from one more unit |
| <b>Indifference curve</b>               | all bundles of equal utility                                             |
| <b>MRS</b>                              | slope of indifference curve = $MU_F / MU_E$                              |
| <b>Budget constraint / line</b>         | $I = p_F F + p_E E$ ; the opportunity set                                |
| <b>MRT</b>                              | slope of budget line = $p_F / p_E$                                       |
| <b>Consumer optimum</b>                 | highest reachable curve tangent to the budget line                       |
| <b>Interior / corner solution</b>       | some of both goods / all of one good                                     |
| <b>Behavioral economics</b>             | systematic deviations from the rational model                            |
| <b>Loss aversion / endowment effect</b> | losses hurt more than equal gains; owning raises valuation               |

## Exam pointers

- **Draw the machine, then solve it.** For "find the optimal bundle": (1) write  $MRS = MU_F / MU_E$  , (2) set  $MRS = p_F / p_E$  , (3) combine with the **budget line**  $I = p_F F + p_E E$  , (4) solve the two equations. *Two equations, two unknowns — always.*
- **Cobb-Douglas → use the shortcut.** If  $U = F^a E^b$  , jump straight to  $F^* = a / (a+b) \cdot I / p_F$  . Saves time; derive it once so you trust it.
- **Check the second good.** After finding one quantity, plug back into the budget line for the other, and **verify it satisfies the constraint** ( $p_F F^* + p_E E^* = I$  ).
- **MRS = MU of the x-axis good over MU of the y-axis good.** Getting the ratio upside-down is the #1 error. If Food is on the x-axis,  $MRS = MU_F / MU_E$  and the tangency is  $MU_F / MU_E = p_F / p_E$  .
- **Corner solutions:** perfect substitutes → compare  $MU_F / p_F$  vs  $MU_E / p_E$  and spend everything on the winner. Perfect complements → ignore the price ratio, use the **fixed ratio + budget line**.
- **Comparative statics language:** income change = **parallel shift**; own-price change = **pivot**. Re-solving the optimum after a price change is how you **derive a demand curve** (Ch. 6).
- **Ordinal, not cardinal:** never say "twice as happy." Utility numbers only **rank**.
- **Behavioral:** be able to name the bias from a scenario (endowment effect, loss aversion, framing, self-control, overconfidence) and say it's a **systematic** (not random) deviation.

**Bridge to Lecture 5 (Ch. 6 — Demand curves & elasticities).** We now have a consumer who picks the tangency bundle for *given* prices and income. **Change one price and re-solve**, over and over, and the optimal quantities trace out a **demand curve** — closing the loop back to the demand we assumed in Lecture 2, and connecting to the **elasticities** of Lecture 3.

## ★ Practice problems (with solutions)

A full-chapter set: preferences & axioms → utility & MU → MRS → budget line & MRT → the optimum → Cobb-Douglas shortcut → corner solutions → comparative statics → behavioral. Try each before opening the solution. Use **Food (F)** on the x-axis, **Entertainment (E)** on the y-axis throughout.

1. A consumer has bundles  $A = (E\ 4, F\ 2)$ ,  $B = (E\ 8, F\ 2)$ ,  $C = (E\ 3, F\ 3)$ ,  $D = (E\ 6, F\ 5)$ . Using only "**more is better**," which rankings can you establish for certain, and which need the indifference map? (Compare A–B, C–D, and A–C.)
2. Preferences satisfy completeness, transitivity, monotonicity. A student says: "*I prefer coffee to tea, tea to cocoa, but cocoa to coffee.*" Which **axiom** does this violate, and why is it a problem for building a utility function?
3.  $U = \sqrt{EF}$ . (a) Find the utility of  $(E\ 2, F\ 8)$ ,  $(E\ 4, F\ 4)$ ,  $(E\ 8, F\ 2)$ ,  $(E\ 5, F\ 5)$ . (b) Which lie on the same indifference curve? (c) Rank all four bundles.
4.  $U = F^{0.5} E^{0.5}$ . (a) Derive  $MU_F$  and  $MU_E$ . (b) Evaluate both at  $(F\ 4, E\ 9)$ . (c) Does  $MU_F$  rise or fall as F increases (E fixed)? Name the principle.
5. For  $U = \sqrt{EF}$ , show that  $MRS = E/F$ . Then compute the MRS at  $(E\ 2, F\ 8)$  and at  $(E\ 8, F\ 2)$  and interpret the difference in one sentence.
6. Income  $I = 40$ ,  $p_F = 4$ ,  $p_E = 2$ . (a) Write the budget line as  $E = \dots$ . (b) Give both intercepts and the slope. (c) Is  $(F\ 6, E\ 8)$  affordable? (d) What is the **MRT**, and what does it mean?
7. Same as 6 ( $I = 40$ ,  $p_F = 4$ ,  $p_E = 2$ ). (a) The government gives a **€20 income top-up**. What happens to the budget line? (b) Instead,  $p_F$  falls to 2. What happens? (c) Sketch the difference in words (shift vs pivot).
8.  $U = \sqrt{EF}$ ,  $I = 40$ ,  $p_F = 4$ ,  $p_E = 2$ . Find the **optimal bundle** using  $MRS = MRT$  + the budget line. Verify it satisfies the constraint.
9.  $U = F^{0.25} E^{0.75}$  (Cobb-Douglas),  $I = 60$ ,  $p_F = 3$ ,  $p_E = 6$ . Use the **expenditure-share shortcut** to find  $F^*$  and  $E^*$ . What share of income goes to each good?
10. (Perfect substitutes.)  $U = 2F + E$ ,  $I = 24$ ,  $p_F = 3$ ,  $p_E = 2$ . (a) Compute  $MU_F/p_F$  and  $MU_E/p_E$ . (b) What is the optimal bundle, and why is it a **corner solution**?
11. (Perfect complements.)  $U = \min(F, E)$ ,  $I = 30$ ,  $p_F = 2$ ,  $p_E = 3$ . Find the optimal bundle. (Hint: the optimum sits at the corner where  $F = E$ ; combine with the budget line.)
12. (Comparative statics.) Start from problem 8's optimum ( $U = \sqrt{EF}$ ,  $I=40$ ,  $p_F=4$ ,  $p_E=2$ ). Now  $p_F$  falls to €2. (a) Find the new optimum. (b) Did the quantity of Food rise or fall? (c) You now have **two  $(p_F, F^*)$  points** — what curve are you tracing out?
13. (Behavioral.) Name the bias in each: (a) You keep a concert ticket you'd never pay today's resale price for. (b) "1 in 3 patients survive" feels better than "2 in 3 die." (c) You put your alarm clock across the

room so you can't hit snooze. (d) 80% of new founders rate their success odds  $\geq 7/10$ .

## Solutions

1. **A vs B:** B has more E (8 vs 4), same F  $\rightarrow B > A$  (more is better). **C vs D:** D has more of *both* (E 6 > 3, F 5 > 3)  $\rightarrow D > C$ . **A vs C:** A has more E (4 > 3) but less F (2 < 3)  $\rightarrow$  **cannot rank** by "more is better" alone; you need the **indifference map** (depends on the MRS).
2. Violates **transitivity**. coffee > tea and tea > cocoa should imply coffee > cocoa, but she says cocoa > coffee — the preferences **cycle**. A cycling ranking **can't be represented by a utility function** (no set of numbers can have coffee > tea > cocoa > coffee), so the whole model of choice breaks down.
3. (a)  $U = \sqrt{EF}$ : (E2,F8)  $\rightarrow \sqrt{16} = 4$ ; (E4,F4)  $\rightarrow \sqrt{16} = 4$ ; (E8,F2)  $\rightarrow \sqrt{16} = 4$ ; (E5,F5)  $\rightarrow \sqrt{25} = 5$ . (b) The **first three** (all  $E \cdot F = 16$ ) lie on the **same indifference curve** ( $U=4$ ). (c) **(E5,F5) > {the other three, which are indifferent to each other}** — i.e.  $(5,5) > (2,8) \sim (4,4) \sim (8,2)$ .
4. (a)  $MU_F = \partial U / \partial F = 0.5 F^{-0.5} E^{0.5} = 0.5 \sqrt{E/F}$ ;  $MU_E = 0.5 F^{0.5} E^{-0.5} = 0.5 \sqrt{F/E}$ . (b) At (F4,E9):  $MU_F = 0.5 \sqrt{9/4} = 0.5 \cdot 1.5 = 0.75$ ;  $MU_E = 0.5 \sqrt{4/9} = 0.5 \cdot 0.667 = 0.333$ . (c) As F rises,  $\sqrt{E/F}$  falls  $\rightarrow MU_F$  falls = **diminishing marginal utility**.
5.  $MRS = MU_F / MU_E = [0.5 \sqrt{E/F}] / [0.5 \sqrt{F/E}] = \sqrt{E/F} \cdot \sqrt{E/F} = E/F$ . At (E2,F8):  $MRS = 2/8 = 0.25$ ; at (E8,F2):  $MRS = 8/2 = 4$ . **Interpretation:** with lots of F she'll give up only 0.25 E for one more F; with little F she'll give up 4 E — the MRS **falls** as F rises (diminishing MRS / convex curve).
6. (a)  $40 = 4F + 2E \rightarrow E = 20 - 2F$ . (b) E-intercept = 20, F-intercept = 10, slope = -2. (c) (F6,E8):  $4 \cdot 6 + 2 \cdot 8 = 24 + 16 = 40 \rightarrow$  **exactly affordable** (on the line). (d)  $MRT = p_F / p_E = 4/2 = 2 \rightarrow$  giving up 1 Food frees €4, which buys 2 Entertainment.
7. (a) **Income +€20** ( $I = 60$ ): **parallel outward shift** — new intercepts  $E = 60/2 = 30$ ,  $F = 60/4 = 15$ ; **slope unchanged** (-2). (b)  **$p_F$  falls to 2:** budget **pivots** — E-intercept **unchanged at 20**, F-intercept rises to  $40/2 = 20$ , slope flattens to  $-p_F / p_E = -1$ . (c) Income change = **parallel shift** (same slope, more of everything); own-price change = **pivot** (one intercept fixed, slope changes).
8.  $MRS = E/F$  (from problem 5). Set  $MRS = MRT$ :  $E/F = 4/2 = 2 \rightarrow E = 2F$ . Budget:  $4F + 2E = 40 \rightarrow 4F + 2(2F) = 40 \rightarrow 8F = 40 \rightarrow F^* = 5, E^* = 10$ . (Check:  $4 \cdot 5 + 2 \cdot 10 = 20 + 20 = 40$  ✓.)
9. Shortcut for  $U = F^{0.25} E^{0.75}$  ( $a=0.25, b=0.75, a+b=1$ ):  $F^* = 0.25 \cdot 60/3 = 0.25 \cdot 20 = 5$ ;  $E^* = 0.75 \cdot 60/6 = 0.75 \cdot 10 = 7.5$ . **Shares:** 25% of income (€15) on Food, 75% (€45) on Entertainment. (Check:  $3 \cdot 5 + 6 \cdot 7.5 = 15 + 45 = 60$  ✓.)
10.  $U = 2F + E$ :  $MU_F = 2$ ,  $MU_E = 1$ . (a)  $MU_F / p_F = 2/3 \approx 0.67$ ;  $MU_E / p_E = 1/2 = 0.5$ . (b) Food gives **more utility per euro** ( $0.67 > 0.5$ ) and the rate is **constant** (perfect substitutes), so spend **everything on Food**:  $F^* = I / p_F = 24/3 = 8$ ,  $E^* = 0$ . **Corner solution** because  $MU_F / p_F \neq MU_E / p_E$  everywhere — there's no interior tangency; the highest indifference (straight) line touches the budget line **on the F-axis**.
11.  $U = \min(F, E)$ : no point in having more of one than the other, so the optimum has  **$F = E$** . Budget:  $2F + 3E = 30$  with  $E = F \rightarrow 2F + 3F = 30 \rightarrow 5F = 30 \rightarrow F^* = E^* = 6$ . (Check:  $2 \cdot 6 + 3 \cdot 6 = 12 + 18 = 30$  ✓.) (Note: the price ratio is irrelevant to the ratio of goods — only to how much of the fixed bundle you can afford.)

12. Now  $p_F = 2$ ,  $p_E = 2$ ,  $I = 40$ .  $MRS = E/F = p_F/p_E = 2/2 = 1 \rightarrow E = F$ . Budget:  $2F + 2E = 40$  with  $E = F \rightarrow 4F = 40 \rightarrow F^* = 10, E^* = 10$ . (b) Food rose from 5 to 10 as its price fell  $4 \rightarrow 2$  (a downward-sloping demand response). (c) The two points ( $p_F=4, F=5$ ) and ( $p_F=2, F=10$ ) are two points on the consumer's demand curve for Food.

13. (a) **Endowment effect** / **sunk-cost + status-quo bias** (you value it more because you own it). (b) **Framing effect** (same facts, "survive" vs "die"). (c) **Lack of self-control** — a commitment device (planner constrains the future doer). (d) **Overconfidence**.

## ★ Fun facts & memorable details (Lecture 4)

Sticky bits from Consumer Choice.

- **"More is better" has a loophole:** economists just **redefine bads as the absence of the bad** — you don't demand pollution, you demand *clean air*. A neat trick that keeps monotonicity intact.
- **Utility is unobservable — so numbers lie.**  $U = 8$  is not "twice as good" as  $U = 4$ ; utility is purely **ordinal**. Early economists agonised over this; the modern answer is "we only ever need the ranking."
- **The bang-for-your-buck rule is the whole optimum.**  $MU_F/p_F = MU_E/p_E$  — at the best bundle, the **last euro** spent on each good buys the **same** extra utility. If it didn't, you'd move money to the better-value good. That's *all* tangency means.
- **Cobb-Douglas people are creatures of habit:** with  $U = F^a E^b$  you always spend the **same share** of income on each good — no matter how rich you get or how prices move. That's why it's the field's favourite functional form.
- **Left shoes are worthless alone.** Perfect complements (L-shaped curves) are why a single earring, one glove, or a lone right shoe adds **zero** utility — you need the matching unit.
- **The ultimatum game breaks "Homo economicus."** Offered a lopsided split, people **pay real money to punish unfairness** — and this holds from Pittsburgh to Papua New Guinea. Fairness is not a Western quirk.
- **You'd sell your coffee mug for far more than you'd pay for it.** The **endowment effect**: Cornell students valued a randomly-given mug **2–4× higher** than students who didn't get one. Ownership rewires valuation.
- **The paradox of voting:** a strict cost-benefit reading says *don't vote* (your vote almost never decides the election). That people vote anyway is a clue that **narrow self-interest is too small a theory of us**.
- **Ulysses tied himself to the mast** to resist the Sirens — a 3,000-year-old **commitment device**, the same logic as not keeping cigarettes in the house. Self-control problems are ancient.